

U.S. ARMY INFORMATION SYSTEMS ENGINEERING COMMAND DRAFTING GUIDE

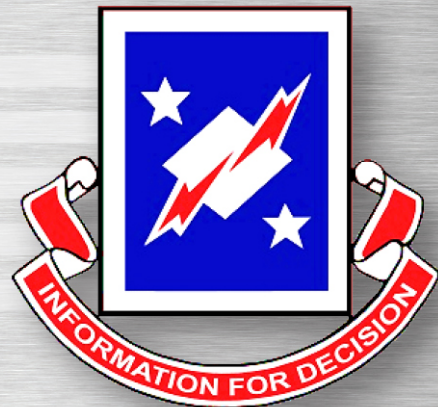
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TECHNICAL DIRECTOR

APRIL 2018

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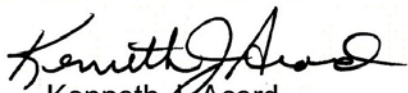
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Kenneth J. Acord
USAISEC Technical Director





EXECUTIVE SUMMARY

This document establishes uniform practices for the preparation and drafting of engineering drawings within the U.S. Army Information Systems Engineering Command (USAISEC). The document is based upon and references the U.S. Army Corps of Engineers (USACE) Architectural, Engineering, and Construction (A/E/C) Computer Aided Design (CAD) Standard (STD), release 6.0 and the National CAD STD (NCS), Version 6.





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U.S. ARMY INFORMATION SYSTEMS ENGINEERING COMMAND DRAFTING GUIDE

1.0 INTRODUCTION

1.1 Background

The U.S. Army Information Systems Engineering Command (USAISEC) developed this drafting guide to provide a baseline for the development of engineering drawings within USAISEC. This guide was developed using material from the U.S. Army Corps of Engineers (USACE) and the National Computer Aided Design (CAD) Standard (STD) (NCS). The intent of the guide is to address drafting practices and processes unique to USAISEC with the Architectural, Engineering, and Construction (A/E/C) CAD STD as the Department of Defense (DoD) STD and the NCS as the foundation.

1.2 Scope

This guide provides basic instructions and recommendations for drafting of engineering drawings created by USAISEC employees and Contractors. The document includes general drafting guidelines for USAISEC. It is understood that this guide cannot address every situation, so if your project or program requires deviation from this guide, a waiver from the USAISEC Technical Director (TD) is required. A Waiver Request Form is available in the TD's SharePoint Library.

2.0 BASIC DRAWING GUIDELINES

2.1 General Guidance

The USACE A/E/C CAD STD is the established standard for the DoD; this guide is meant to provide additional clarification or address USAISEC unique drafting practices. Many of the guidelines provided in the following pages are also provided on the Standard Drawing (STD)-MS-000A, which is shown in Appendix H of this guide. STD-MS-000A will be referenced throughout this guide. The electronic version of this drawing is available in the TD's SharePoint library. Many other helpful documents including border templates, are available in this library.

Many engineers and technicians have AutoCAD on their computers and are producing project engineering drawings. This guide provides a recommended standardized format and scheme for engineering drawings produced for USAISEC. Use of this guide will save time for the draftsman and will allow for standardization on drawings. Some long term programs have developed specific processes, which may continue to be used. Guidance for specific programs is identified in the appendices of this guide. Drawings can be provided to the draftsman in any format to include hand sketches, Visio, PowerPoint, and AutoCAD/AutoCAD Lt. AutoCAD/AutoCAD Lt is the preferred method of submission. All drawings will be returned from the draftsman in AutoCAD.

2.2 Types of Drawings

There are five basic types of drawings commonly used within USAISEC. The five basic types of drawings are:



a. Engineering Drawing. Drawings developed for the design and installation of equipment and systems. These are drawings for specific projects that will be implemented in the future.

b. Redline Drawing. Drawings that have been redlined/updated to reflect the actual method or location of installation (after completion) of a specific project. These drawings are delivered to a draftsman, so As-Built drawings can be developed. The drawings can be engineering drawings or standard drawings that have been redlined/updated.

c. As-Built. Drawings which reflect the actual method or location of installation (after completion) of a specific project. These drawings are usually the result of engineering drawings which have been annotated during installation to show changes necessitated by on-site conditions. The drawings are typically delivered to the customer as a finished product.

d. STD. STDs contain information that can be used (and reused) by design and installation personnel on various facilities and projects to design, specify, or install specific equipment or systems. These drawings are not specific to any particular site.

e. Plant-in-Place. Drawings used as a record of what was installed and how it was installed. Plant-in-Place drawings are drawings which reflect the current configuration of an operating communications facility. The Plant-in-Place drawings differ from as-built drawings since they cover the whole facility not just a particular installation or upgrade.

2.3 Orientation

a. Plans, elevations, and details. Views shall be oriented on the sheet, so that elevations and features are aligned whenever possible. When detailing is being added, details shall appear on the sheet based on their orientation on the feature. For instance, a detail of the top of a wall shall be orientated above a detail of the bottom of the wall. If a detail is taken from a large-scale plan or elevation, the orientation shall remain the same as the view from which the cut was taken. If this is not possible, a note stating that the orientation was changed will be added (i.e., VIEW ROTATED 90-DEGREES). When two or more plans of the same structure or the plans of two or more different structures are put on the same drawing, the orientation of all must conform to one another and to their relative positions on the ground.

b. Maps and drawings. Civil works maps and drawings shall be oriented, so that north is toward the top of the sheet when practicable, or toward the left of the sheet, if top orientation is impracticable.

c. Vicinity maps. Vicinity maps will be oriented with north toward the top of the sheet.

2.4 Line Work

To be consistent with the USACE A/E/C CAD STD, Tables 3-1, 3-2, 3-3, and the NCS, Section 4.2.7, various types of lines require specific line weights.

2.5 Fractions and Commas/Spacing in Dimensions.

Scaling has to be exact to the actual dimensions. For metric units (per the A/E/C CAD STD) or inch-pound units, a space replaces the traditional comma in numbers containing five or more digits (see Figure 1). In numbers containing four digits, no space is





necessary. Dimensions can be either inline or above the line, and must remain consistent for all drawings within a project.

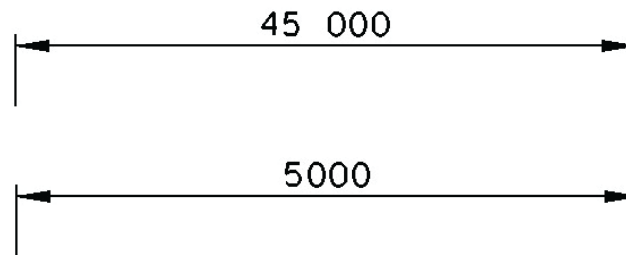


Figure 1. Format for Dimensions

2.6 Symbolology

The library of common symbols is listed in Appendix D of the USACE A/E/C CAD STD. Project specific symbols are listed in the border sheet model files.

a. Symbol descriptions shall follow the following basic guidelines to allow their repetition from job to job, preclude conflicts within the drawings, and maximize suitability for CAD interface:

1. Symbol descriptions in legends shall be concise and worded in a comma-separated structure that flows from the general to the specific. For example, in lieu of wording the description of a convenience receptacle as “WALL-MOUNTED, DUPLEX RECEPTACLE,” structure the wording as “RECEPTACLE, DUPLEX, WALL-MOUNTED.”

2. The symbol description should not contain specific information that would normally be contained in equipment schedules or on the plans.

3. Where equipment or devices within a room are all identical, one symbol can be identified with the full type identifier and other information, and a callout can be referenced to this symbol with the following words: “Typical for this room.” Do not use the word “typical” by itself without additional wording to show where the typicality applies.

b. Symbol identifiers. Identifying letters within and around symbols must be legible at one-half normal size. Therefore, letter sizes within and around symbols shall be a minimum of 1/8-inch (3-millimeters (mm)) in height (unless the NCS specifically calls out a text height of 3/32-inch (2.5-mm) for that symbol (e.g., Detail Indicator)).

2.7 Redlining

The development of redlined drawings takes place during the installation using a standard method of denotation and process. Drawings that have been redlined/updated, reflect the actual method of installation (after completion). Although some USAISEC programs/projects have specific processes in place for redlining drawings within AutoCAD, the following general guidelines shall be used for denotation:

- Red will indicate changes
- Yellow highlight will indicate deletions
- Blue will indicate notes to the draftsman





The redlined drawings will be used to develop the as-built drawings. The as-built drawings from the project are combined at the site/facility to the Plant-in-Place drawing repository. The Plant-in-Place drawings include all of the as-built drawings from all previous installations/upgrades at the site/facility.

2.8 Drawing Revisions

Drawing practices associated with drawing changes shall be consistent with those already used on the drawing to be revised, unless the latest of applicable standards can be incorporated without conflict. Any change to a drawing after release shall be recorded in the Revision History Block.

2.8.1 Revision Methods

Changes may be made by adding, deleting, crossing out the information, or by re-drawing the drawing.

2.8.2 Re-Drawn Drawings

Depending on the circumstances, a drawing may be redrawn either with or without change.

2.8.3 With Change (Revision Letter is Advanced)

When a revision warrants re-drawing of the drawing, the revision letter next in sequence shall be entered in the REV column of the Revision History Block of the new drawing. See Figure 2.

2.8.4 Without Change (Advance of Revision Letter is not Required)

When a drawing, or a sheet of a drawing is to be replaced because of loss, destruction, or degradation due to age, it may be re-drawn without change. The replacement shall duplicate the requirements of the old original, which is being replaced.

2

1

REVISION				
ZONE	REV	DESCRIPTION	DATE	APPROVED
	X	COMPUTER DRAWN, AUTOCAD 13	07OCT13	
	A	MINOR REVISIONS	07OCT13	

Figure 2. Example Revision Block

2.9 Superseding a Drawing with that of a Different Number

2.9.1 Superseding (New) Drawing

When a drawing is superseded by a drawing with a different number, the notation "REPLACES WITH CHANGE DRAWING (Enter superseded drawing number) REV





(Enter superseded revision letter, or if no revision, enter a – [dash])” shall be entered in the DESCRIPTION columns of the superseding drawing.

No entry is required in the REV letter, DATE, or APPROVED Columns. New Title Block Approval entries apply.

2.9.2 Superseding (Old) Drawing

When the superseded drawing will be retained, the notation “REPLACED WITH CHANGE BY DRAWING (Enter the superseding drawing number) REV (Enter the revision letter of the superseding drawing, or if no revision, enter a – (dash))” shall be entered in the DESCRIPTION column of the superseded drawing.

2.10 Revisions of Digital Data

Revisions of digital data files and copies of digital data files shall not be considered re-draws, except when the document is converted from a manually maintained to a digitally maintained document.

2.10.1 Revision Letters

- Upper case letters shall be used in sequence beginning with “A” and omitting letters “I”, “O”, “Q”, “S”, “X”, and “Z”.
- When the single letters have been exhausted, the revisions following “Y” shall be “AA” through “AY”.
- Should “AA” to “AY” be exhausted the next sequence shall be “BA” through “BY”.
- Revision letters shall not exceed two characters.
- Initial issue of a drawing does not constitute the need for a revision letter and may be indicated by the use of a - (dash).

2.10.2 Revision Locations

When incorporating a change or replacing the drawing, a revision location shall be identified using:

- Description in the Revision History Block
- Zone locations in the ZONE Column of the Revision History Block

2.10.3 Multiple Changes.

All changes authorized by a single revision request shall be incorporated into the document at the same time. All changes to a drawing incorporated at one time shall be identified by the same revision letter, if the revision letter is assigned at the time the changes are incorporated.

Sequence Number Use: When a revision involves two or more individual changes on a drawing, each change may be identified by a sequence number enclosed in parentheses preceding the description of the change.

2.11 Drawing Packages Organization and Index

Drawing packages shall be organized using the drawing index sequence shown in Table 1 and on Figure D-2 (STD-MS-000A, sheet 2) in Appendix D of this document.



**Table 1. Drawing Index Sequence**

DRAWING INDEX SEQUENCE			
DETAIL CODE	DESCRIPTION	DETAIL CODE	DESCRIPTION
DI	DRAWING INDEX	FD	FABRICATION DETAILS
EL	MAJOR EQUIPMENT LIST	AT	ANTENNA DETAILS
LM	LIST OF MATERIALS	ID	INTERCONNECT DIAGRAM
SY	SYSTEM DIAGRAM	WD	WIRING DIAGRAM
MP	MULTIPLEX PLAN	CL	CABLE RUNNING LIST
TD	TRUNKING DIAGRAM	CR	CABLE ROUTING
TS	TRUNKING SCHEMATIC	CN	CABLE CONNECT LIST
VM	VICINITY MAP	CS	CABLE CONNECTOR PANEL
BA	BASE LAYOUT	CA	CIRCUIT AND CHANNEL ASSIGNMENT
SP	SITE PLAN	ST	STRAPPING DIAGRAM
AE	ARCHITECTURAL AND ENGINEERING	XC	CROSS CONNECT DIAGRAM
BL	BUILDING LAYOUT	BC	BLOCK CABLE RECORDS
CV	CABLE VAULT	BK	BLOCK AND LEVEL DIAGRAM
OP	OUTSIDE PLANT	MG	MIXING AND GRADING PLAN
MH	MAINTAINENCE HOLE DETAILS	MS	MISCELLANEOUS DETAILS
FP	FLOOR PLAN	PB	PATCH BAY
RE	RACK ELEVATION	RA	RESTRICTED AREA
WE	WALL ELEVATION	RP	RADIATION PROFILE
DF	DISTRIBUTING FRAME	SC	SCHEMATIC DIAGRAM
TB	TERMINAL BLOCK LAYOUT AND ASSIGNMENT	TN	NETWORK
TE	TERMINAL STRIP ASSIGNMENT	TM	TELEPHONE MISCELLANEOUS
RD	CABLE RACK DIAGRAM	AD	ASSEMBLY DRAWING
GS	GROUNDING SYSTEM	CD	COMPUTER DESK
AS	ALARM SYSTEM	DT	DIGITAL TAPE
PD	POWER DIAGRAM	EF	EQUIPMENT FACE LAYOUT
LP	LIGHTING PLAN	PH	PHOTOGRAPH
AC	AIR CONDITIONING PLAN	TA	AUDIO VISUAL
AX	AUXILIARY FRAME	VT	VIDEO TAPE
IN	INSTALLATION DETAILS		





2.11.1 Text in Drawings

2.11.2 General Notes

Notes on drawings shall be clear and concise. General Notes are notes with universal application to work on all drawings or to all work on specific drawings. General Notes shall be worded such that they are independent of the drawing(s), without cross-referencing or pointing with leader arrows to plans, details, etc. General Notes shall be capable of being removed from the drawings and placed in the specifications. The title above a group of General Notes applicable to sheets for a particular discipline (e.g., Architectural, Electrical, Mechanical, etc.) shall be worded "GENERAL NOTES: APPLICABLE TO ALL (Insert Discipline) SHEETS." General Notes applicable to specific sheets shall be worded "GENERAL NOTES: THIS SHEET ONLY" or "GENERAL NOTES: SHEETS _____ THRU _____." Notes shall not include contractual requirements, such as statements of costs; time and place of delivery; methods of payment; and requirements for submission, approval, or distribution of data or reports. All General Notes will be placed in the far right column of the drawing area of the Sheet File. Wherever General Notes are used, they will be shown with a 1/4-inch (6-mm) title heading of "GENERAL NOTES." All notes under the General Notes heading will be numbered sequentially starting with 1.

2.11.3 Subject Titles

When only one subject appears on a drawing, and its title appears in the title block, a subtitle should be shown under each view or detail on the drawing. When more than one subject appears on a drawing, a title should be shown under each subject, and a subtitle shown below each view or detail in that group.

2.11.4 Orientation

Text shall be set parallel to the primary base of the drawing. If necessary, text can be rotated at 30-degree angles up to 180-degrees, as long as the orientation is as shown in Figure 3. However, rotating the text is discouraged to prevent having to turn the drawing sheet to read notations.

Note: An exception to maintaining this text orientation would be on waterway projects because of the various directions in which channels are located. Often, text that has a definite bearing on the contract is kept at proper orientation, while map features incidental to the contract are allowed to follow the orientation when created in a north-up base map, which may result in upside-down text on rotated plan sheets.



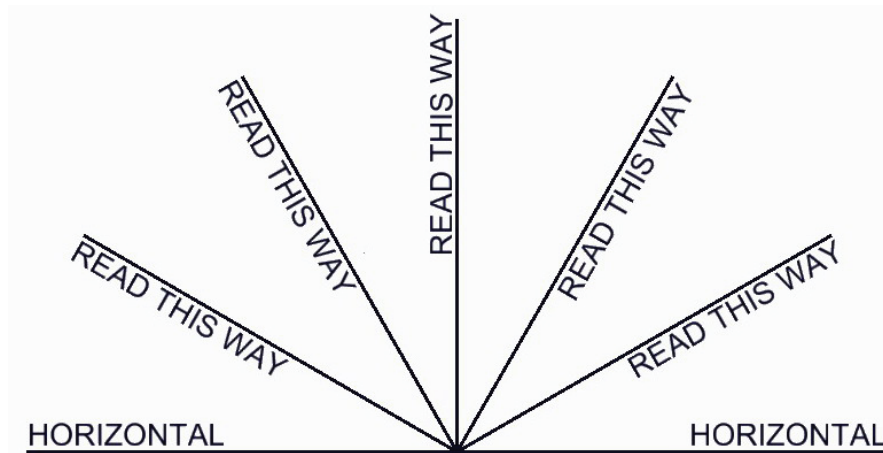


Figure 3. Orientation of Text

2.11.5 Abbreviations

Abbreviations for words or phrases frequently used in plans, sections, elevations, or details should follow discipline-unique abbreviations. Abbreviations for words or phrases frequently used in plans, sections, elevations, or details should follow the abbreviations as established in the A/E/C CAD STD and NCS. A copy of the NCS abbreviations list is available in the TD's SharePoint library. In addition, the following rules regarding abbreviations shall be followed:

- Abbreviations shall be used only when necessary and when their meanings are unquestionably clear. When in doubt, spell out.
- Once an abbreviation has been used, the same abbreviation must be continued throughout the project document.
- Only one abbreviation is allowed for each nomenclature.
- Any abbreviation used shall be identified in the abbreviations list of the drawing set.

3.0 DRAWING FILE ORGANIZATION

3.1 Model Space

All drawings will be done in model space. Model space drawings (model files) represent full-size drawings of building elements, systems, or information (e.g., the mechanical Heating, Ventilation, and Air Conditioning (HVAC) system, the architectural floor plan, details, or sections), and sheet files represent final plotted sheets. Model files are used as components in creating plotted sheet files. The information contained within a model file for a discipline may be referenced by other disciplines to create the particular model files or sheet files for that discipline.

a. Model-space-specific information represents the physical form of a site, a building, or other elements. This information is often shared between CAD files (both model space and paper space) through the use of external reference files. Examples include walls, doors, light fixtures, racking, cabinets, and equipment. Model-space-specific information may be either literal (e.g., walls) or symbolic (e.g., electrical outlets).





b. Layout-specific information may include notes, annotative symbols, and titles. This type of information is usually not shared between CAD files or drawings. Design models inside a layout contain graphic information that would relate to real-world information (e.g., point coordinates), or information that would be sectioned off into multiple sheets (e.g., a floor plan that may take three sheets to present because of its size). Layout-specific information would include items specific for the presentation of that sheet. This is one reason that layouts should never be used as a reference file to other files.

3.2 Layout Space

Layouts are the final project sheets that are ready to be plotted. A layout contains specific information (e.g., north arrows, scales, section cuts, title block information) in a layout (Paper Space). A design model inside the layout contains the model space information assembled as it would be displayed in paper space. This model would have real-world spatial alignment, and would be used as the primary model for graphical information to be displayed and presented in the layout.

3.3 List of Materials (LOM)

The LOM required for the assembly of items in a drawing shall be listed, as shown on STD-MS-000A (see Appendix D). Note that abbreviations are allowed in the LOM (this is an exception to paragraph 2.11.5). The listing shall include the following:

- a. Find Number. The column FIND NO. is a sequential number used to identify the items contained in the LOM. The number is used in conjunction with a circle and pointing arrow in the drawing, which identifies the items location. The number is used to cross-reference an item on the list with the location of the item in the drawing.
- b. Description. This is a short description of the item.
- c. Manufacturer. This is the manufacturer or source of the item.
- d. Part Number. This is the manufacturer's part number or National Stock Number (NSN) (if required) for the item.
- e. Unit of Issue (UI). This is the UI of the material. An example would be Feet (FT) for a length of cable.
- f. Quantity (QTY). This is the QTY required of the item for this particular assembly or on this particular drawing.

The condensed STD-MS-000A drawing shown in Figure 4, provides examples of both the lower right hand corner Title Block and the Title Block aligned vertically along the right side of the drawing. Either method is acceptable and must remain constant throughout all drawings on a project.





4	DELETED (.0450)				
3	A TRIPLE LINE ENTRY IS A .25 IN SPACE WHICH IS .0625 ADDED TO .1875 AN ENTRY OF 4 LINES WOULD BE .3375 IN, ETC	16882K	5310-00-616-2793	EA	1
2	A DOUBLE LINE ENTRY IS A .1875 IN SPACE ADD .0625 FOR EACH ADDITIONAL LINE	09141J	5310-00-809-3079	EA	1
1	A SINGLE LINE ENTRY IS A .125 IN SPACE	07177Y	5306-00-151-2068	EA	1
FIND NO	DESCRIPTION	MANUFACTURER	PART NO/NSN	UI	QTY

PARTS LIST (.0675)									
IDENT NO (.0775) STD-MS-000A					US ARMY INFORMATION SYSTEMS ENGINEERING COMMAND (.0675)				
SHEET 1 OF 2 (.0450)					FORT HUACHUCA (.0575) ARIZONA				
DESIGNED BY ENGINEER (.0450) DATE 07OCT13					COMPUTER-AIDED DRAFTING STANDARDS (.0775)				
DRAWN BY DRAFTER (.0450) DATE 07OCT13									
CHECKED BY DATE									
APPROVED BY APPROPRIATE CSE DATE									
DESIGN ACTIVITY ELIE-ISE-xx (.0450)					DRAWING NO 50470				
DWG INDEX NO: xxxxxx-Dixxxxxx (.0450)					SCALE NONE (.0450) 1" SHEET 1 OF 2				

Figure 4. STD-MS-000A Title Block

3.4 Drawing Numbers and Electronic File Naming Conventions (USAISEC Modification from the USACE A/E/C STD)

The naming conventions are the same for the drawing numbers and the electronic drawing files. The scheme described below allows CAD users to determine the contents of a drawing without actually displaying the file. It also provides a convenient and clear structure for organizing drawing files within project directories. The following is a recommended approach for naming drawing files.

The model file naming convention (Figure 5) is broken into three parts. The first part has four mandatory fields. A hyphen is used to separate the three parts. The second part has two mandatory fields. The third part has two fields and is used for revision control of the drawing. All fields must be filled out to correctly identify the location, type of facility, the drawing type, and the revision for the drawing.



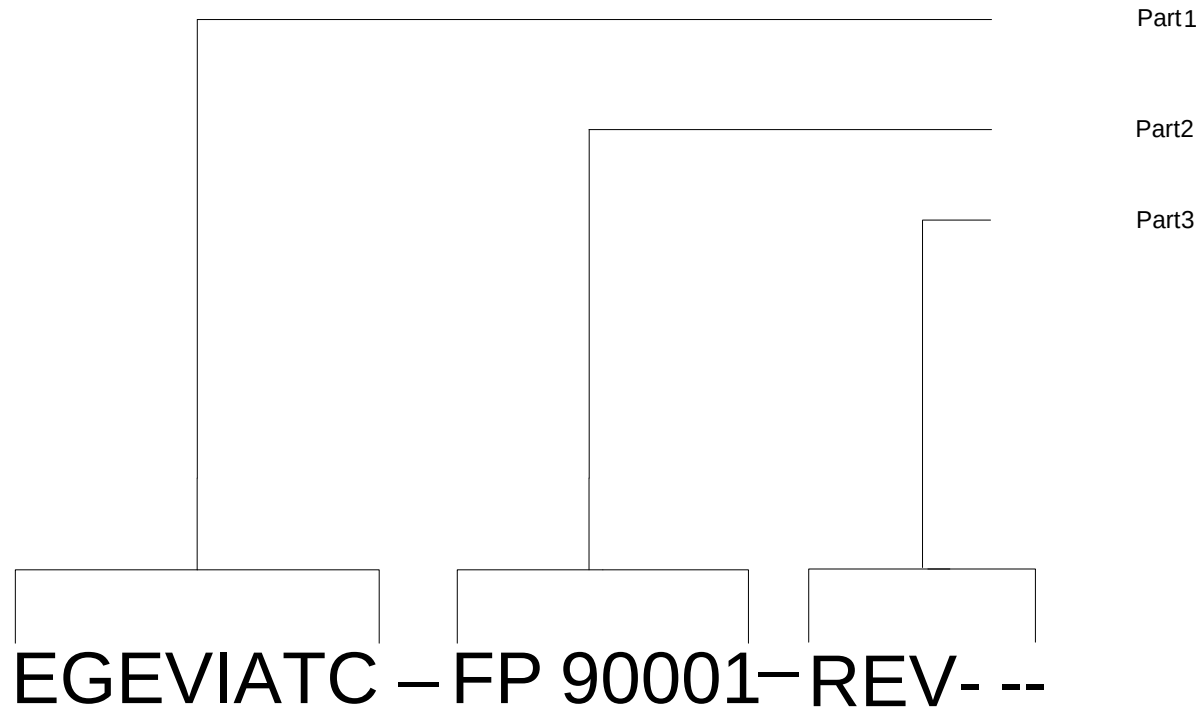


Figure 5. Project Drawing File Naming Convention

Part one of the naming convention is as follows (See Figure 6):

- The first field is a one character field for the geographic location or theater of operation that the facility is physically located. Examples: E-Europe, C-Continental United States (CONUS), S-Southwest Asia (SWA), P-Pacific. Refer to Appendix B Table B-2 for a list of this information.
- The second field is a two character country or state in which the facility is physically located. Examples: GE-Germany, BE-Belgium, AZ-Arizona, KO-Korea. Refer to Appendix B Table B-3 for a list of this information.
- The third field is a three character field abbreviation for the name of the facility. Examples: VIA-Patch Barracks, FTH-Fort Huachuca. Refer to Appendix B Table B-4.
- The fourth field is a two character field that identifies the type of facility. Examples: TC-Technical Control Facility, TL-Telephone Switch, MW-Microwave. Refer to Appendix B Table B-4 for a list of this information.

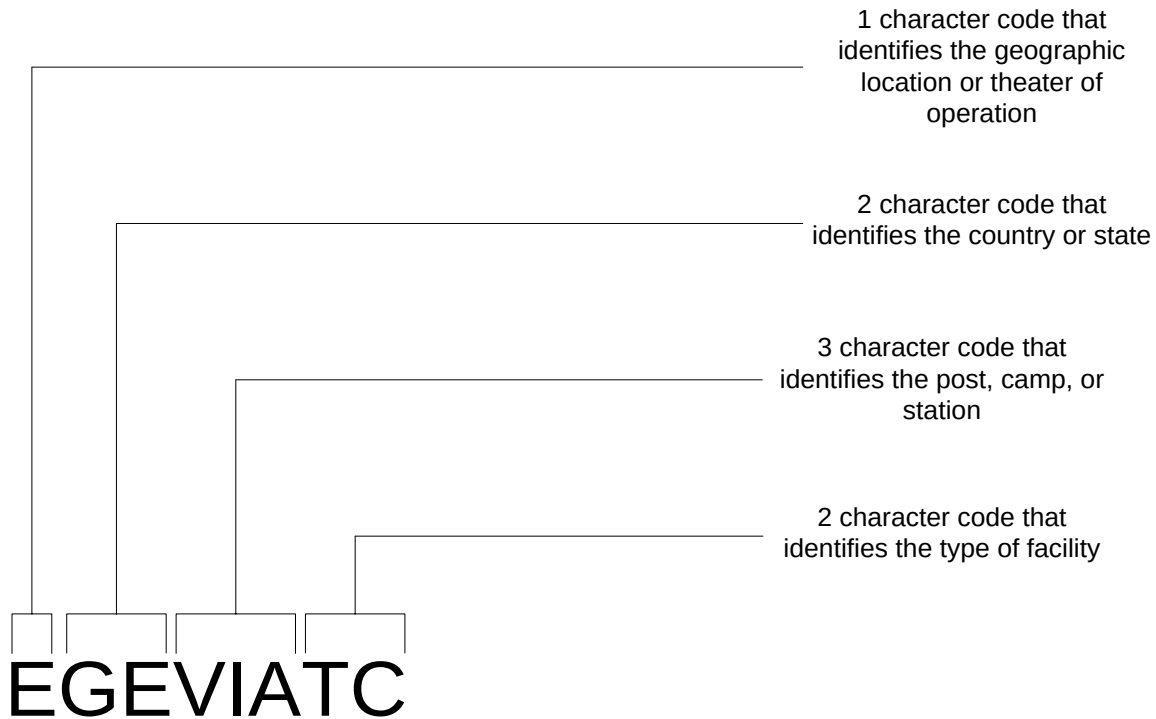


Figure 6. Part One of File Naming Convention

Part two of the naming convention is as follows (See Figure 7):

- The first field is a two character field to describe the type of drawing. Examples: FP-Floor Plan, RE-Rack Elevation, RD-Cable Rack Diagram. Refer to Appendix E for a list of this information.
- The last field is a five position numeric designator. Use a numeric sequence beginning with 00001. Drawing numbers assigned to project drawings at USAISEC will have a “9” in the first position, i.e., 90001. Drawing numbers assigned to drawings prepared by USAISEC Contractors will have an “8” in the first position.

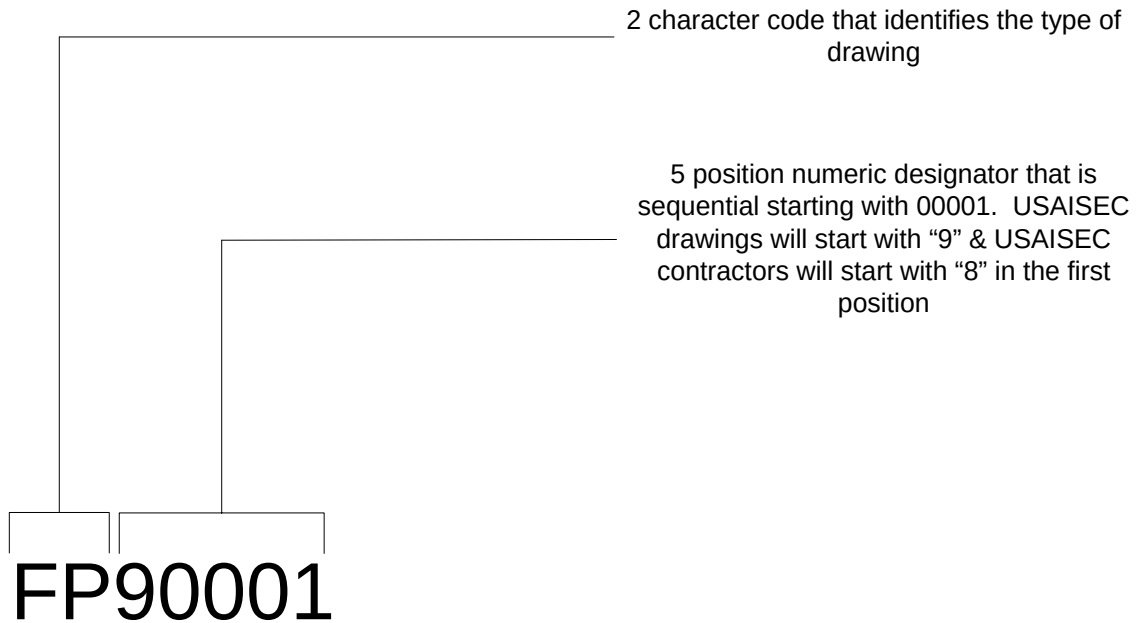


Figure 7. Part Two of File Naming Convention

Part three of the naming convention is as follows (See Figure 8):

- The first field is a three character field to describe the revision of drawing. This will always be REV.
- The last field is a two position character designator. Use a character sequence beginning with "A" for the first revision to an original drawing file.
- Upper case letters shall be used in sequence beginning with "A" and omitting letters "I", "O", "Q", "S", "X", and "Z".
- When the single letters have been exhausted, the revisions following "Y" shall be "AA" through "AY".
- Should "AA" to "AY" be exhausted, the next sequence shall be "BA through "BY".
- Revision letters shall not exceed two characters.
- Initial issue of a drawing does not constitute need for a revision letter and may be indicated by the use of a - (dash).



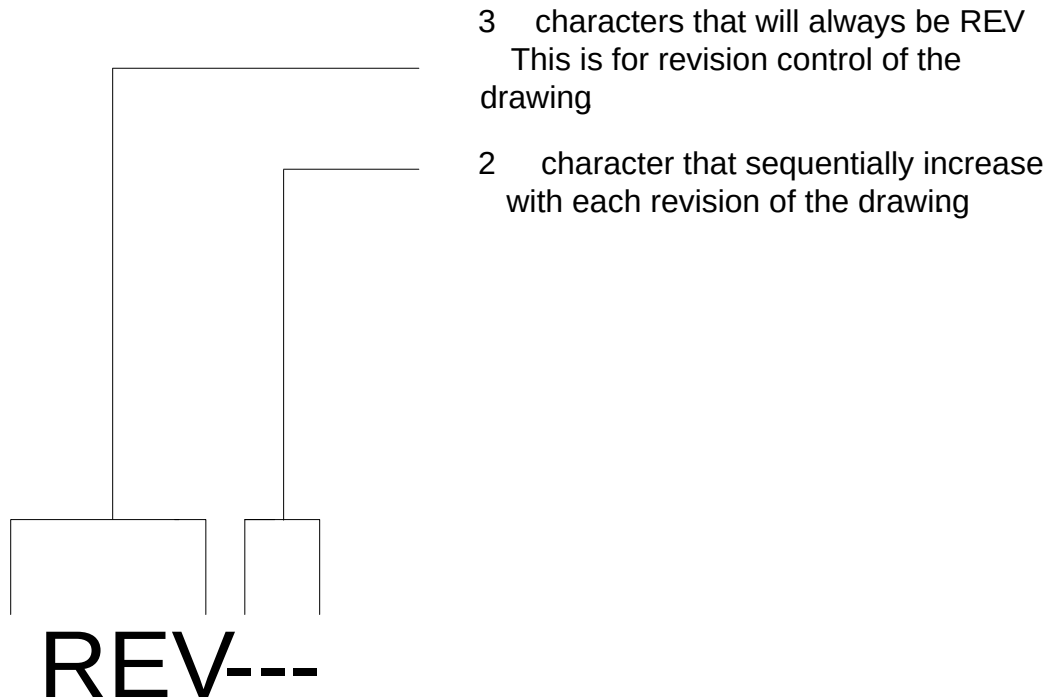


Figure 8. Part Three of File naming Convention

3.5 USAISEC Common Drawing Designators

The following list indicates the general order of drawings in a drawing package; however, not all drawings are necessary for each project. The drawing package starts with an index. The index should include columns for a drawing number, revision number, title, and remarks. The following descriptions provide information on the common USAISEC drawings.

a. Drawing Index “DI”. This is the first sheet or drawing in each set. This is a complete list of drawings for a project, site, or facility. The drawing index is usually prepared on an American National Standards Institute (ANSI) B-size sheet (see Table 2 for description of sizes) or on several sheets, and it lists all of the drawings in the set by drawing number and title. This index shows the number of sheets for each drawing if more than one sheet is used, and it displays the revision of each drawing. Usually, only one drawing index is prepared for a site or facility; however, when the site is part of several major communications systems, more than one may be required. If a particular drawing is applicable to more than one system because of positioning, interconnects, etc., it will be listed on each of the respective drawing indexes.

b. System Diagram “SY”. This drawing shows the interrelationships and connectivity of the component elements of a system. This diagram uses labeled circles or rectangles to represent system elements and uses lines to represent connections. This diagram also uses symbols to portray special characteristics of system components (i.e., repeaters, passive reflectors, interfaces with other systems, etc.).

c. Base Layout Drawing “BA”. This is a to-scale drawing that provides the base boundaries, including the placement of properties, roads, streets, railroads, waterways,





runways, and survey markers, as well as the location of the different Communications-Electronics (C-E) facilities. This drawing inserts a vicinity map to illustrate the physical relationship of the installation to the surrounding area. In addition, the base layout drawing proportions the vicinity map to size with the to-scale drawing. The installation layout drawing may contain a grid overlay to serve as the key sheet for Outside Plant (OSP) drawings. The grid should divide the base into rectangles that are 1,600-feet high and 2,400-feet wide. If it is necessary to show the details of a congested area on subsequent OSP drawings, the grid should use a larger scale, such as 800-feet by 1,200-feet. Grid lines should not cut primary buildings or C-E facilities. If this is not possible, do not divide these salient features into more than two parts. Details of each grid section will be contained on large-scale OSP drawings.

d. Site Plan “SP”. This drawing shows the overall site plan of the C-E facility and displays a plan view of the installation, to include the location and dimensions of major components and the location of survey markers, antennas, and antenna azimuths. A site plan may show manholes, ducts, underground and buried cables, aerial cables, and other OSP facilities, when not provided on base grid or cable diagram drawings.

e. Architectural and Engineering Drawing “AE”. These drawings show the design and engineering required for the manufacture of specific items, such as structures, buildings, open areas, or equipment. Architectural engineering drawings encompass construction requirements, manufacturing specifications, material requirements, style, character, and aesthetic values. The information is definitive, precise, and to-scale. The USACE can often provide this information for new buildings.

f. Building Layout Drawing “BL”. This is the key drawing for the family of drawings pertaining to a specific building. A building layout drawing is required for each building containing C-E facilities or portions thereof. This drawing must identify the room number and/or operational use of each room or area. The drawing notes state the type of building construction, floor composition, walls, and ceiling, as well as the floor capacity, ceiling heights, area floor space, and miscellaneous information. The plan view for this drawing is normally a single-line building outline showing all interior walls and partitions, and the plan view may show power ducts. The building layout drawing should be a to-scale drawing using as many sheets and match lines as necessary. Usually, this drawing requires one sheet for each floor. In this drawing, single lines represent inside walls, and breaks in the lines represent internal and external access openings of the building. Furthermore, symbols indicate electric, power, water, heat, and ventilation service entrances or internal source locations in a building layout drawing. This drawing identifies the major C-E facility and equipment locations, the supporting floor plans, and the location of the building as shown on the base layout map.

g. Cable Vault Drawing “CV”. This drawing shows the location, layout, racking, cabling, splicing, steps, access ports, and ceiling sleeves in a cable vault. This drawing also shows the relationship of the vault to the building and an elevation view showing length, width, headroom, and duct arrangement. In addition, a cable vault drawing shows the arrangement of cables on racks and the approximate location of valves, pressure plugs, sleeves, and splices, as well how the cables enter the vertical side of the Main Distribution Frame (MDF). This drawing must identify the cable or trunk number, cable





count, and number of pairs. Prepare only one cable vault drawing per building with C-E functions.

h. Floor Plan “FP”. Use this drawing to support a building layout drawing. A floor plan provides detailed information about specific communication rooms. A floor plan is drawn with the maximum space available at the top of the sheet to leave room for an equipment legend and description. The drawing title must contain the numerical identification of the building, room number, and area, as well as the base and/or site name and address. A floor plan must contain a plan view of a room or area as identified by the building layout drawing. A floor plan also must show wall thicknesses, door swing directions, and access openings, as well as the location of all partitions, columns, and dimensions, to establish special limits in all directions. A floor plan identifies interior dimensions, numerical designations, and/or operational uses of all rooms or areas. Note any element of restriction to the installation, maintenance, or operation of C-E equipment on this drawing. Identify all C-E equipment installed or programmed onsite, and ensure the equipment is shown to scale and identified in the drawing legend. Heavier lines should indicate equipment fronts in a floor plan.

i. Rack Elevation Drawing “RE”. This is a dimensional, elevation view of equipment racks, consoles, and cabinets showing the equipment arrangement and blank panels. In this drawing, the item number must identify equipment when an equipment legend is used. A rack elevation drawing must show the shapes and sizes of compartments, assignments of space, and general mechanical and electrical details of component mounting. Note the building, room, equipment row, and operational use of the equipment on this drawing. Prepare one rack elevation drawing for each facility.

j. Wall Elevation Drawing “WE”. This drawing depicts all equipment. This includes furniture and accessories mounted on transportable van and shelter walls or on building walls.

k. Distribution Frame (DF) Drawing “DF”. This drawing shows the placement, layout, arrangement, and numbering of each MDF, Intermediate Distribution Frame (IDF), and Combined Distribution Frame (CDF). This drawing must identify the terminal blocks and cable run numbers of all cable runs to the frame, as well as terminal block designations. This drawing also includes the location and identification of protectors and terminal strips on a frame, whether these are in use or reserved for future use. This type of drawing should indicate set jacks and detail views presented when those terminal strips serve more than one circuit. Only one set of DF drawings is normally prepared for each building. Each sheet of the drawing will show only one DF.

l. Terminal Block Layout and Assignment Drawing “TB”. The installers and maintenance personnel use these drawings to identify each terminal block in a DF (MDF, IDF, CDF, etc.), each terminal block’s function, and where the other end of the cables terminate or originate. This data will include pictorial views of the blocks and frame, identification of the blocks, and pertinent information in narrative or tabular form. This drawing’s data will be used in conjunction with the Cable Running List (CRL). The terminal block layout and assignment drawing should identify all spare blocks, terminals, wires, and cables.

m. Cable Rack Diagram “RD”. This drawing shows the construction and placement of cable racks, runways, raceways, trays, ducts, etc., used to support the interfacing





cables. This diagram will use a plan view that shows the cable rack relative to equipment and building features. This diagram will also use an elevation view to supplement the plan view by using heavy lines to indicate the rack and light lines to indicate the equipment. A cable rack diagram should identify the type and size of components in the drawing (i.e., conduits, elbows, dropouts, and reducers), if engineering requirements demand such information.

n. Grounding System Drawing “GS”. These drawings show all the details necessary to depict all grounding systems associated with the facility, both inside and outside the building. Schematically, these drawings also show how and where cable shields, equipment cabinets, individual equipment, and other devices are grounded and bonded. Grounding system drawings show the entire communication facility’s grounding system, including onsite power distribution, protective ground, lightning protective ground, signal ground, special Electromagnetic Pulse (EMP) protective ground, and an Earth Electrode Subsystem (EES). In addition, these drawings show all specialized hardware and techniques used to provide grounding, such as the use of chemicals, counterpoises, ground rods, wells, etc. Grounding system drawings should also display the EES’s resistance-to-earth measurements and the methods of measurement, including the date of the last measurement.

o. Power Diagram “PD”. These drawings are specific in nature to identify power and the corresponding circuits. These diagrams are one-line diagrams that describe the Alternating Current (AC) and Direct Current (DC) power systems, emergency and standby power, and safety/utility grounds (but not signal grounds), including special grounds installed for lightning and EMP protection. Power diagrams show all circuits, breakers, conduits, and other related hardware available for use in the power system. These diagrams also show power distribution to the equipment and cabinet(s), including the circuit assignments and power capacities of all outlets. In addition, power diagrams show unique features, such as Uninterruptible Power Supply (UPS) systems, automatic switchover systems, and monitor and control circuits. These diagrams should clearly identify any provisions required to expand the power system. Furthermore, power diagrams should include any tabular and narrative data necessary to describe the entire power system clearly.

p. Unistrut Plan “UP”. This drawing will provide a plan view of the unistrut installed in the technical equipment area. A unistrut plan is a superimposed drawing on the technical area rack layout. This plan provides a detailed view depicting the unistrut cross-section and model number (i.e., P-1000).

q. Installation Drawing “IN”. These drawings show how to mount, install, or place individual equipment at the facility. In addition, installation drawings provide details concerning the specific locations and actual securing of items. These drawings also provide supplemental information that is not available in the standard installation practice documents/drawings or manufacturer’s documents.

r. Interconnect Diagram “ID”. This block diagram shows the electrical connections of a system, as well as its components, parts, and devices. This diagram also shows the actual interconnection of equipment when it differs from the functional interconnection shown on the block and level diagrams. This detailed drawing will assist in the development of the CRL. The more accurate and detailed the information is on an





interconnect diagram, the easier it will be to develop the CRL. An interconnect diagram helps to ensure the CRL is not missing any cables or interconnections between equipment.

s. CRL “CL”. These lists assist in identifying, in tabular form, the origin and termination of all cables used in the facility (i.e., signal, power, protective ground, signal ground, radio frequency cables and waveguides, control, and intercom). Symbols and narrative data supplement CRLs when necessary. The information shown in this list includes cable types, cable and wire sizes, cable functions, circuit assignments, identification coding, unused and spare wires and cables, and any special characteristics of the cables and wires. Use CRLs in conjunction with interconnect diagrams. The CRL will refer to standards and/or installation details that will tell the installer where to find a diagram for the cable assembly. This reference will include the drawing number, sheet number, and/or detail and table as required.

t. Circuit and Channel Assignment Drawing “CA”. This tabular drawing designates circuits and/or channel assignments to communications equipment that process more than one channel (i.e., patch panels, consoles controlling several radios, etc.). A diagram of the face of the equipment may assist in telling the installer what information to place on the label and where to place that label on the equipment being installed.

u. Cross-connect Diagram “XC”. This drawing depicts all permanent and semi-permanent wiring cross-connections between equipment on the distribution frame. This diagram includes the identification and location of equipment racks that interconnect.

v. Miscellaneous Detail Drawing “MS”. Use this drawing when requiring specific information that cannot be categorized in the detail codes of any other drawing listed in this document.

w. STD Drawing “SD”. This is a common and repeatedly used drawing showing the preferred configurations of installation items. This drawing contains information used by installation personnel to install specific equipment or systems.

x. Audiovisual (A/V) Drawing “TA”. Use this drawing for details specific to A/V equipment and systems.

y. OSP Drawing “OP”. This drawing shows the OSP, including buried cable, underground conduit, terminals, poles, and other features similar to OSP facilities. These features include cable numbers, number of pairs, pair counts, and terminal counts. Usually, this drawing is keyed to the grid system established by the base layout.

z. Maintenance Hole (MH) Detail Drawing “MH”. This drawing shows the information required to install underground cables in a specific manhole. This drawing should provide a description of the manhole, to include the number and dimensions, location of entry points, cable racks, and type of drainage. In addition, this drawing should contain the number, type, material, and inside diameter of conduits; the arrangements of cable; and the location of coil cases and other pertinent equipment for manhole use.





4.0 BORDER SHEETS

4.1 Templates

Templates for USAISEC Border sheets are shown in Appendix E. Note that continuation sheets have a different border than sheet 1 (Figure E-1). The electronic version of the USAISEC Drawing Templates are located in the TD's SharePoint Library.

4.2 Sheet Sizes

Typical USAISEC projects (including contract documents) will be drawn on D-size and printed on ANSI B-size sheets. Other industry standard sizes may be used depending on project requirements. Table 2 lists the standard sizes of all sheets.

Table 2. ANSI, Architectural, and International Organization for Standardization (ISO) Sheet Size Information

ANSI, Architectural, and ISO Sheet Size Comparison					
ANSI		Architectural		ISO	
Mark	Size in inches	Mark	Size in inches	Mark	Size in inches (mm)
F	28.0 X 40.0	F	30.0 X 42.0	N/A	N/A
E	34.0 X 44.0	E	36.0 X 48.0	A0	33.1 X 46.8 (841 X 1189 mm)
D	22.0 X 34.0	D	24.0 X 26.0	A1	23.4 X 33.1 (594 X 841 mm)
C	17.0 X 22.0	C	18.0 X 24.0	A2	16.5 X 23.4 (420 X 594 mm)
B	11.0 X 17.0	B	12.0 X 18.0	A3	11.7 X 16.5 (297 X 420 mm)
A	8.5 X 11.0	A	9.0 X 12.0	A4	8.3 X 11.7 (210 X 297 mm)

N/A=Not Applicable

5.0 LAYER ASSIGNMENTS

CAD layers serve to separate graphic elements (lines, shapes, and text) according to the design discipline they represent. Table 3 shows the typical model layers that will be used in USAISEC telecommunications projects. Table 4 shows the layers that will be used in the Layout Space layers. A complete listing of layers is identified in the NCS Version 6.

5.1 Model Space Layers

Model files represent full-size drawings of building elements, systems, or information (e.g., the mechanical HVAC system, the architectural floor plan, telecommunication systems, details, or sections), and sheet files represent final plotted sheets. Drawings in Model Space contain information in various layers. This provides structure to the AutoCAD file and simplifies the exchange of information. A recommended layering scheme for telecommunication systems is shown in Table 3. This scheme follows the NCS layer guidelines for naming layers with a discipline, a major group, up to optional minor groups, and an optional status field to clearly identify each layer. An exhaustive listing of standard layer naming details is available in the NCS Appendix A, which is located in the TD's SharePoint Library.



**Table 3. Telecommunication Model Space Layers**

Layer	Description
T-ANNO-DIMS	Witness/extension lines, dimension terminators, and dimension text.
T-ANNO-KEYN	Reference keynotes with associated leaders.
T-ANNO-NOTE	General notes and remarks.
T-ANNO-NPLT	Non-plotting graphic information.
T-ANNO-PATT	Patterning, poche, shading, and hatching.
T-ANNO-RDME	Read-me information.
T-ANNO-SYMB	Miscellaneous symbols.
T-ANNO-TEXT	Miscellaneous text and callouts with associated leaders.
T-ANNO-REFR	An AutoCAD user-specific layer for use in attachment of external references (i.e., reference files).
TA-	Audiovisual.
TC-	Clock and Program.
TD-	Telecommunications Demolition
TE-CABT	Equipment Enclosures i.e., Equipment Cabinets.
TE-RERA	Equipment Enclosures Relay Racks.
TE-DISF	Equipment Enclosures Distribution Frames.
TG-SIGR	Telecommunications Grounding Signal Reference Ground.
TG-DCRF	Telecommunications Grounding DC Reference Ground.
TG-PWDC	Telecommunications Grounding DC Pathway Ground.
TI-	Intercom and public address systems.
TM-	Monitoring and alarm systems.
TN-	Network cabling and equipment.
TP-SIGC	Telecommunications Pathway Signal Pathway (Copper).
TP-SIGF	Telecommunications Pathway Signal Pathway (Fiber Optic).
TP-PWRA	Telecommunications Pathway AC Power Pathway.
TP-PWRD	Telecommunications Pathway DC Power Pathway.
TS-	SCADA Supervisory Control and Data Acquisition (SCADA) systems and equipment.
TT-	Telephone systems, wiring, and equipment.
TY-	Security Access Control and alarm systems.

5.2 Layout (Paper Space) Layers

Layouts are the final project sheets that are ready to be plotted. A layout contains specific information (e.g., north arrows, scales, section cuts, title block information) in a layout (Paper Space). This information shall be contained in the appropriate layers. Table 4 provides a recommended approach to Paper Space Annotation Layer names and functions.



**Table 4. Layout (Paper Space) Layers**

Layer	Description
T-ANNO-DIMS	Sheet-specific witness/extension lines, dimension terminators, and dimension text.
T-ANNO-KEYN	Sheet-specific keynotes with associated leaders.
T-ANNO-LEGN	Legends and symbol keys.
T-ANNO-NOTE	Sheet-specific notes and general remarks.
T-ANNO-NPLT	Non-plotting graphic information.
T-ANNO-PATT	Sheet-specific patterning and hatching (e.g., keyplan patterning).
T-ANNO-RDME	Read-me information.
T-ANNO-REDL	Redlines.
T-ANNO-REVS	Revisions, amendments, addenda, and modifications.
T-ANNO-SCHD	Schedules.
T-ANNO-SYMB	Sheet-specific symbols (e.g., north arrow, scales).
T-ANNO-TEXT	Sheet-specific text and callouts with associated leaders.
T-ANNO-REFR	An AutoCAD user-specific layer for use in attachment of external references (i.e., reference files).

6.0 AUTOCAD SYMBOLS AND BLOCKS

6.1 Symbols

Symbols are defined as a graphic representation of an object or material (see example in Figure 9). Common symbols used by USAISEC and Industry are available in the TD SharePoint Library. The symbols in the library are based on the NCS Symbols (1200 plus (+) symbols) with USAISEC specific symbols included:



PROPOSED SPLICE SYMBOL
(ONE SYMBOL = ALL SPLICE CASES)

Figure 9. Example Symbol for a Proposed Cable Splice

6.2 Blocks

USAISEC drawings will utilize AutoCAD Blocks where feasible. The AutoCAD block is defined as one or more objects combined to create a single object (see Figure 10). Common blocks used by USAISEC and Industry are available in the TD's SharePoint Library.





Figure 10. Example Block for Modernization of Enterprise Terminals (MET) Rack Type 4

7.0 STD DRAWINGS

STD drawings are defined as drawings that are prepared for projects that have repeated usage and show the preferred configurations. The drawings are prepared and issued by USAISEC and will be identified as STD Drawings by the number which appears in the identification block in the STD Title Block. All STD Engineering Implementation Plan (EIP) drawings are STD drawings.

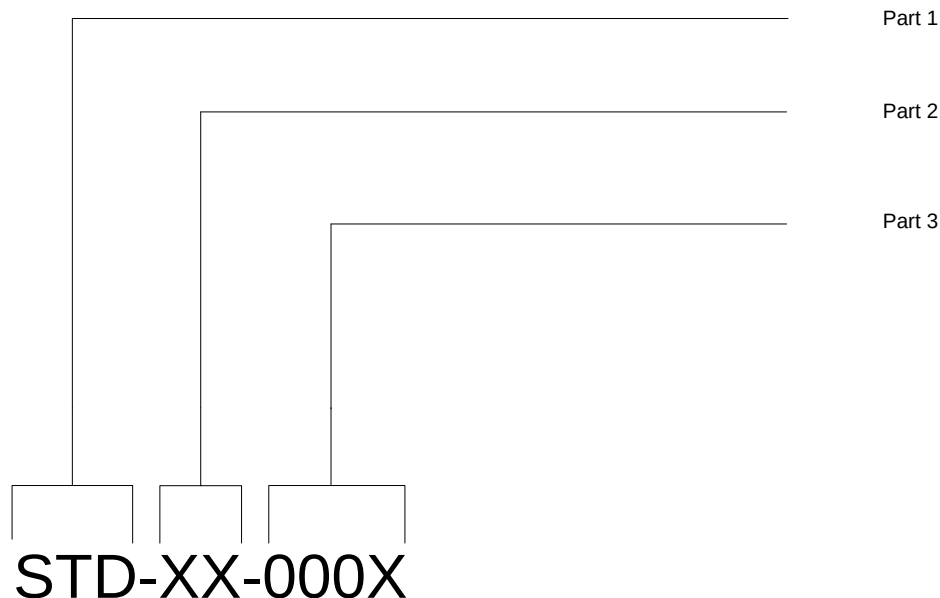


Figure 11. STD Drawing Naming Convention

Parts one through three of the naming convention are as follows:

- The first field is a three character field that will always contain the characters “STD”. This will identify the drawing as a USAISEC STD drawing.
- The second field is a two character field that represents the Functional Area that the drawing represents. The characters will be taken from Table 5 below.



- The third field is a four character field that will be assigned by the Project Lead (PL).

Table 5. Standard Drawing Functional Areas

CODE	DESCRIPTION
AS	Alarm System.
BB	Broadband Communications Systems.
CM	Common Applications and Configuration Managed drawings.
CA	Carrier Systems.
DS	Data Switching Systems (routers, switches, matrix, ATM, fiber switches).
DW	Dense Wave Division Multiplexing Systems.
HF	HF Radio.
IN	Inside Plant.
MP	Multi-Protocol Label Switching.
MS	Miscellaneous.
MX	Multiplexer.
MW	Microwave.
OP	Outside Plant.
PT	Patch and Test Facility.
RG	Range Communications.
SA	Space Allocation drawings.
SC	Single Channel Non-switched Communications System.
TA	Audiovisual.
SS	Satellite Communications System.
TC	Multifunction Technical Control Facility.
TF	Training Facility.
TR	Tropospheric Scatter Radio Communications System.
WD	Wiring Diagrams.
XP	Transportable Communications System.





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APPENDIX A. REFERENCES

The following publications are applicable to this document herein.

A-1.0 U.S. GOVERNMENT (USG) PUBLICATIONS

- a. USACE, A/E/C CAD Standard, release 6, September 2012.

A-2.0 NON-USG PUBLICATIONS

- a. American Society of Mechanical Engineers (ASME), ASME Y14.100-2013, Engineering Drawing Practices, 2013.
- b. National Institute of Building Sciences, National CAD Standard Version 6, No date.





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APPENDIX B. TRANSMISSION SYSTEM ENGINEERING DRAWINGS

B-1.0 TOBYHANNA ARMY DEPOT (TYAD) DRAWINGS

This first section is provided for reference only. TYAD has a layering convention and redlining convention that differs from the standard methods presented in this guide. Table B-1 shows an example of the layer naming scheme used by TYAD for Wideband Satellite projects. Note the numbers contained in the parenthesis correspond to drawing tabs and also to scheduled equipment updates.

Table B-1. TYAD Layering Convention

Layer	NAMING SCHEME
1	(1) AS BUILT Misc
2	(1) AS BUILT Panduit ADD
3	(1) AS BUILT Plenums
4	(1) AS BUILT Power Conduit
5	(1) AS BUILT Racks
6	(1) AS BUILT REDLINES ADD
7	(1) AS BUILT REDLINES REM
8	(1) AS BUILT Tiles
9	(1) AS BUILT Trays
10	(1) AS BUILT Under Floor Cable Tray
11	(1) AS BUILT Walls
12	(2) UPDATE Future Rack ADD
13	(2) UPDATE Future Rack REM
14	(2) UPDATE Misc ADD
15	(2) UPDATE Misc REM

B-2.0 TROJAN PROGRAM DRAWING NUMBERS

The drawing numbers for TROJAN are issued by the Program Manager (PM) at Research, Development, and Engineering Communications (RDECOM). The PM has issued a series of drawing numbers to be used by USAISEC, and the numbers are managed by the TROJAN team.

B-3.0 DRAWING IDENTIFICATION (ID) NUMBERING CODES

Table B-2 identifies the geographic location one-character ID code.

Table B-2. Geographic Location One-Character ID Code

Geographic Location	
Geographic ID Code	Location
A	Africa
C	CONUS
E	Europe
P	Pacific
S	Southwest Asia





Table B-3 identifies the country/state location and two-character ID code.

Table B-3. Country/State Location and Two-Character ID Code

Country		
Geographic Location	Country	Country/State ID Code
CONUS	United States	AK
CONUS	United States	AL
CONUS	United States	AR
CONUS	United States	AZ
CONUS	United States	CA
CONUS	United States	CO
CONUS	United States	CT
CONUS	United States	DE
CONUS	United States	FL
CONUS	United States	GA
CONUS	United States	HI
CONUS	United States	IA
CONUS	United States	ID
CONUS	United States	IL
CONUS	United States	IN
CONUS	United States	KS
CONUS	United States	KY
CONUS	United States	LA
CONUS	United States	MA
CONUS	United States	MD
CONUS	United States	ME
CONUS	United States	MI
CONUS	United States	MN
CONUS	United States	MO
CONUS	United States	MS
CONUS	United States	MT
CONUS	United States	NC
CONUS	United States	ND
CONUS	United States	NE
CONUS	United States	NH
CONUS	United States	NJ
CONUS	United States	NM
CONUS	United States	NV
CONUS	United States	NY
CONUS	United States	OH
CONUS	United States	OK
CONUS	United States	OR
CONUS	United States	PA
CONUS	United States	RI
CONUS	United States	SC
CONUS	United States	SD
CONUS	United States	TN
CONUS	United States	TX
CONUS	United States	UT
CONUS	United States	VA
CONUS	United States	VT
CONUS	United States	WA



**Table B-3. Country/State Location and Two-Character ID Code (continued)**

Country		
Geographic Location	Country	Country/State ID Code
CONUS	United States	WI
CONUS	United States	WV
CONUS	United States	WY
Europe	Belgium	BE
Europe	Bosnia	BO
Europe	Bulgaria	BU
Europe	Czech Republic	CZ
Europe	Germany	GE
Europe	Italy	IT
Europe	Netherlands	NL
Europe	Poland	PO
Europe	Romania	RO
Europe	Slovakia	SL
Europe	Spain	SP
Europe	United Kingdom	UK
Pacific	Australia	AU
Pacific	Guam	GU
Pacific	Japan	JA
Pacific	Korea	KO
Pacific	Okinawa	JO
Southwest Asia	Afghanistan	AF
Southwest Asia	Bahrain	BA
Southwest Asia	Iraq	IQ
Southwest Asia	Jordan	JR
Southwest Asia	Kuwait	KU

Table B-4 identifies the site name and three-character ID code.

Table B-4. Site Name and Three-Character ID Code

Sites		
Geographic Area/Country/Code	Site Code	Site Name
Europe / Belgium / BE	CHE	Chievres
Europe / Belgium / BE	FLO	Flobecq
Europe / Belgium / BE	HOU	Houtem
Europe / Belgium / BE	LEC	Le Chenol
Europe / Belgium / BE	SPB	SHAPE
Europe / Belgium / BE	SPA	Spa Ma Lachamps
Europe / Belgium / BE	WES	Westrozebeke
Europe / Italy / IT	AVO	Aviano
Europe / Italy / IT	CEG	Ceggia
Europe / Italy / IT	CIM	Cimagallina
Europe / Italy / IT	CLO	Coltano
Europe / Italy / IT	MTC	Mt Cimone
Europe / Italy / IT	MCA	Mt Corna
Europe / Italy / IT	MTS	Mt Serra
Europe / Italy / IT	MTE	Mt Venda
Europe / Italy / IT	PAG	Paganella
Europe / Italy / IT	VCA	Vicenza



**Table B-4. Site Name and Three-Character ID Code (continued)**

Sites		
Geographic Area/Country/Code	Site Code	Site Name
Europe / Germany / GE	ANU	Adenau
Europe / Germany / GE	BKH	Bad Kreuznach
Europe / Germany / GE	BAN	Bann
Europe / Germany / GE	BHR	Baumholder
Europe / Germany / GE	BIG	Bitburg
Europe / Germany / GE	BST	Bonstetten
Europe / Germany / GE	BDH	Brandhof
Europe / Germany / GE	BTL	Breitsol
Europe / Germany / GE	CBL	Campbell Barracks Heidelberg
Europe / Germany / GE	DNG	Darby Nuernberg
Europe / Germany / GE	DST	Darmstadt
Europe / Germany / GE	DON	Donnersberg
Europe / Germany / GE	FEL	Feldberg
Europe / Germany / GE	FRI	Friolsheim
Europe / Germany / GE	GAB	Gablingen
Europe / Germany / GE	GEG	Giegenwang
Europe / Germany / GE	GSN	Giessen
Europe / Germany / GE	GRA	Grafenwoehr
Europe / Germany / GE	HAN	Hanau
Europe / Germany / GE	HDM	Heidenheim
Europe / Germany / GE	HFL	Hohenfels
Europe / Germany / GE	HST	Hohenstadt
Europe / Germany / GE	KLN	Kaiserslautern
Europe / Germany / GE	KAT	Katterbach
Europe / Germany / GE	KSL	Koenigstuhl
Europe / Germany / GE	LDL	Landstuhl
Europe / Germany / GE	LKF	Langerkopf
Europe / Germany / GE	MHN	Mannheim
Europe / Germany / GE	MEL	Melibocus
Europe / Germany / GE	MHL	Muhl
Europe / Germany / GE	PKS	Patch Barracks Vaihingen
Europe / Germany / GE	PMS	Pirmasens
Europe / Germany / GE	RSN	Ramstein
Europe / Germany / GE	RAG	Reese Augsburg
Europe / Germany / GE	RMN	Rhein Main
Europe / Germany / GE	RBS	Robinson Barracks Stuttgart
Europe / Germany / GE	SCH	Schoenfeld
Europe / Germany / GE	SCW	Schwetzingen
Europe / Germany / GE	SHE	Sembach
Europe / Germany / GE	SPM	Spangdahlem
Europe / Germany / GE	SSB	Stocksberg
Europe / Germany / GE	WBN	Wiesbaden
Europe / Germany / GE	WBG	Wuerzburg
Europe / Germany / GE	ZUG	Zugspitze
Europe / United Kingdom / UK	BRD	Barford St John
Europe / United Kingdom / UK	BKY	Barkway
Europe / United Kingdom / UK	BFM	Botley Hill Farms
Europe / United Kingdom / UK	BOV	Bovington
Europe / United Kingdom / UK	CHV	Chelveston





Table B-4. Site Name and Three-Character ID Code (continued)

Sites		
Geographic Area/Country/Code	Site Code	Site Name
Europe / United Kingdom / UK	CRS	Christmas Commons
Europe / United Kingdom / UK	CDW	Cold Blow
Europe / United Kingdom / UK	CRO	Croughton
Europe / United Kingdom / UK	DAV	Daventry
Europe / United Kingdom / UK	DNK	Dunkirk
Europe / United Kingdom / UK	GID	Goonhilly Downs
Europe / United Kingdom / UK	HEY	High Wycombe
Europe / United Kingdom / UK	HIN	Hillingdon
Europe / United Kingdom / UK	LAK	Lakenheath
Europe / United Kingdom / UK	MEN	Menwith Hill
Europe / United Kingdom / UK	MIL	Mildenhall
Europe / United Kingdom / UK	LDN	Navy London
Europe / United Kingdom / UK	SNG	Swingate
Pacific / Okinawa / JO	CSB	Camp Schwab
Southwest Asia / Kuwait / KU	AFJ	Camp Arifjan
Pacific / Korea / KO	CRC	Camp Red Cloud
Europe / Belgium / BE	BNA	Ben Ahin
Pacific / Okinawa / JO	CHN	Camp Hansen
Pacific / Okinawa / JO	WHB	White Beach
Pacific / Okinawa / JO	AWS	Awase
Pacific / Okinawa / JO	CCY	Camp Courtney
Pacific / Okinawa / JO	BLP	Bolo Point
Pacific / Okinawa / JO	YAK	Yaedake
Pacific / Okinawa / JO	FTB	Fort Buckner
Pacific / Okinawa / JO	KAD	Kadena
Pacific / Okinawa / JO	FMA	Futenma
Pacific / Okinawa / JO	CPK	Camp Kinser
Pacific / Okinawa / JO	NAH	Naha
Pacific / Okinawa / JO	IES	Ie Shima
Pacific / Okinawa / JO	FOS	Camp Foster
Pacific / Okinawa / JO	LZO	LZO
Pacific / Okinawa / JO	NTA	Northern Training Area
Pacific / Okinawa / JO	HFF	Hill 450
Pacific / Okinawa / JO	SBE	Torii Station
CONUS / United States / AZ	FTH	Fort Huachuca TSEEF





Table B-5 identifies the types of facilities.

Table B-5. Types of Facilities

Facility Types	
Facility Code	Description
MS	Miscellaneous
MW	Microwave
PT	Patch and Test Facility
SS	Satellite Communications System
TC	Technical Control Facility
TF	Training Facility
TL	Telephone Switch
XP	Transportable Communications System

Table B-6 provides a listing of Terrestrial Technical Control (TTC) site codes.

Table B-6. Listing of TTC Site Codes

Site Code	Site Name	Facility Type
CMDFTDTC	Fort Detrick	Technical Control Facility
CAZFTHTF	Fort Huachuca TSEEF	Training Facility
EBEBNATC	Ben Ahin	Technical Control Facility
EBECHETC	Chievres	Technical Control Facility
EBEFLOT	Flobecq	Technical Control Facility
EBEHOUTC	Houtem	Technical Control Facility
EBELECTC	Le Chenol	Technical Control Facility
EBESPBTC	SHAPE	Technical Control Facility
EBESPATC	Spa Ma Lachamps	Technical Control Facility
EBEWESTC	Westrozebeke	Technical Control Facility
EGEANUTC	Adenau	Technical Control Facility
EGEBKHTC	Bad Kreuznach	Technical Control Facility
EGEBANTC	Bann	Technical Control Facility
EGEBHRTC	Baumholder	Technical Control Facility
EGEBIGTC	Bitburg	Technical Control Facility
EGEBSTTC	Bonstetten	Technical Control Facility

**Table B-6. Listing of TTC Site Codes (continued)**

Site Code	Site Name	Facility Type
EGEBDHTC	Brandhof	Technical Control Facility
EGEBTLTC	Breitsol	Technical Control Facility
EGECBLTC	Campbell Barracks Heidelberg	Technical Control Facility
EGEDNGTC	Darby Nuernberg	Technical Control Facility
EGEDSTTC	Darmstadt	Technical Control Facility
EGEDONTC	Donnersberg	Technical Control Facility
EGEFELTC	Feldberg	Technical Control Facility
EGEFRITC	Friolsheim	Technical Control Facility
EGEGABTC	Gablingen	Technical Control Facility
EGEGEGTC	Giegenwang	Technical Control Facility
EGEGSNTC	Giessen	Technical Control Facility
EGEGRATC	Grafenwoehr	Technical Control Facility
EGEHANTC	Hanau	Technical Control Facility
EGEHDMTC	Heidenheim	Technical Control Facility
EGEHFLTC	Hohenfels	Technical Control Facility
EGEHSTTC	Hohenstadt	Technical Control Facility
EGEKLNTC	Kaiserslautern	Technical Control Facility
EGEKATTC	Katterbach	Technical Control Facility
EGEKSLTC	Koenigstuhl	Technical Control Facility
EGELDLTC	Landstuhl	Technical Control Facility
EGELKFTC	Langerkopf	Technical Control Facility
EGEMHNTC	Mannheim	Technical Control Facility
EGEMELTC	Melibocus	Technical Control Facility
EGEMHLTC	Muhl	Technical Control Facility
EGEPKSTC	Patch Barracks Viahingen	Technical Control Facility



**Table B-6. Listing of TTC Site Codes (continued)**

Site Code	Site Name	Facility Type
EGEPMSTC	Pirmasens	Technical Control Facility
EGERSNTC	Ramstein	Technical Control Facility
EGERAGTC	Reese Augsburg	Technical Control Facility
EGERMNTC	Rhein Main	Technical Control Facility
EGERBSTC	Robinson Barracks Stuttgart	Technical Control Facility
EGESCHTC	Schoenfeld	Technical Control Facility
EGESCWTC	Schwetzingen	Technical Control Facility
EGESHETC	Sembach	Technical Control Facility
EGESPMTC	Spangdahlem	Technical Control Facility
EGESSBTC	Stocksberg	Technical Control Facility
EGEWBNTC	Wiesbaden	Technical Control Facility
EGEWBGTC	Wuerzburg	Technical Control Facility
EGEZUGTC	Zugspitze	Technical Control Facility
EITAVOTC	Aviano	Technical Control Facility
EITCEGTC	Ceggia	Technical Control Facility
EITCIMTC	Cimagallina	Technical Control Facility
EITCLOTC	Coltano	Technical Control Facility
EITMTCTC	Mt Cimone	Technical Control Facility
EITMCATC	Mt Corna	Technical Control Facility
EITMTSTC	Mt Serra	Technical Control Facility
EITMTETC	Mt Venda	Technical Control Facility
EITPAGTC	Paganella	Technical Control Facility
EITVCATC	Vicenza	Technical Control Facility
EUKBRDTC	Barford St John	Technical Control Facility
EUKBKYTC	Barkway	Technical Control Facility



**Table B-6. Listing of TTC Site Codes (continued)**

Site Code	Site Name	Facility Type
EUKBFMTC	Botley Hill Farms	Technical Control Facility
EUKBOVTC	Bovington	Technical Control Facility
EUKCHVTC	Chelveston	Technical Control Facility
EUKCRSTC	Christmas Commons	Technical Control Facility
EUKCDWTC	Cold Blow	Technical Control Facility
EUKCROTC	Croughton	Technical Control Facility
EUKDAVTC	Daventry	Technical Control Facility
EUKDNKTC	Dunkirk	Technical Control Facility
EUKGIDTC	Goonhilly Downs	Technical Control Facility
EUKHEYTC	High Wycombe	Technical Control Facility
EUKHINTC	Hillingdon	Technical Control Facility
EUKLAKTC	Lakenheath	Technical Control Facility
EUKMENTC	Menwith Hill	Technical Control Facility
EUKMILTC	Mildenhall	Technical Control Facility
EUKLDNTC	Navy London	Technical Control Facility
EUKSNGTC	Swingate	Technical Control Facility
PKOCRCTC	Camp Red Cloud	Technical Control Facility
PJOAWSTC	Awase	Technical Control Facility
PJOBLPTC	Bolo Point	Technical Control Facility
PJOCCYTC	Camp Courtney	Technical Control Facility
PJOFOSTC	Camp Foster	Technical Control Facility
PJOCHNTC	Camp Hansen	Technical Control Facility
PJOCPKTC	Camp Kinser	Technical Control Facility
PJOCSBTC	Camp Schwab	Technical Control Facility
PJOFTBTC	Fort Buckner	Technical Control Facility



**Table B-6. Listing of TTC Site Codes (continued)**

Site Code	Site Name	Facility Type
PJOFMATC	Futenma	Technical Control Facility
PJOHFFTC	Hill 450	Technical Control Facility
PJOIESTC	Ie Shima	Technical Control Facility
PJOKADTC	Kadena	Technical Control Facility
PJOLZOTC	LZO	Technical Control Facility
PJONAHTC	Naha	Technical Control Facility
PJONTATC	Northern Training Area	Technical Control Facility
PJOSBETC	Torii Station	Technical Control Facility
PJOWHBTC	White Beach	Technical Control Facility
PJOYAKTC	Yaedake	Technical Control Facility
SKUAFJTC	Camp Arifjan	Technical Control Facility
SKUAFJTL	Camp Arifjan	Telephone Switch





APPENDIX C. A/V AND VISUAL INFORMATION SYSTEM (VIS) DRAWINGS

C-1.0 A/V LAYER NAMING CONVENTION

The purpose of implementing this convention is to consolidate the layers that are associated in order to expedite identifying and locating them in AutoCAD. This enables the user to easily turn associated layers on or off to obtain specialized views of the systems without having to create multiple detail drawings to separate the systems in model space. All the systems can be drawn in one place in model space connecting to all relevant devices (which are only drawn once) and separate layouts created to isolate each system for better clarity. (Because all the systems shown on a single drawing becomes very busy and potentially confusing or hard to read.)

NOTE: Most letter group codes are, and should be taken directly from NCS naming convention, as much as possible.

a. First Group – Discipline. Recommend always utilizing two letters from the NCS guide to designate the discipline for the layer; such as TA for Telecommunications A/V, TN for Telecommunications (Ethernet) Network, EP for Electrical Power, etc.

b. Second Group – Major System. The next four letter group identifies the type of system. For networks, this would be the name or identity of the network such as Non-classified Internet Protocol Router Network (NIPRNet), Secret Internet Protocol Router Network (SIPRNet), CENX for generic Combined Enterprise Regional Information Exchange System (CENTRIXS), CNXK for CENTRIXS-Korea (K), etc. For A/V systems, it is recommended to identify the audio, video, and control portions separate as SAUD, SVID, and CTRL respectively. There may be a potential issue in differentiating between a video only-type connection and a combined A/V connection. It is suggested that SVID be used for the video only-type connections and that TVVS (Television and Video System from NCS) be used for combined A/V connections (e.g., standard Cable Television (CATV), feeds to displays with audio, etc.).

c. Third Group – Subsystem. The next four letter group would identify the function within that system; such as EQPM for Equipment, CABL for copper cables, FIBR for Fiber Optic Cable (FOC), etc.

It is undetermined at this time, whether to recommend using a fourth group to differentiate between functions such as analog and digital signals (ANLG for analog cables, DIGT for digital, SNGL for Single Mode (SM) fiber, MULT for Multimode (MM) fiber, etc.), or creating a special four letter code in the third group (CANL for analog cables, CDIG for digital cables, FBRS for SM fiber, FBRM for MM fiber, etc.).





Table C-1 identifies suggestions for layer names which illustrate the concept.

Table C-1. Layer Names

NAME	DESCRIPTION
TA-CTRL-ANLG	ANALOG OR SERIAL CONTROL
TA-CTRL-DIGT	DIGITAL CONTROL LINKS (NOT ETHERNET)
TA-CTRL-EQPM	CONTROL EQUIPMENT
TA-SAUD-ANLG	ANALOG AUDIO
TA-SAUD-DIGT	DIGITAL AUDIO
TA-SAUD-EQPM	AUDIO EQUIPMENT
TA-SVID-ANLG	ANALOG VIDEO (NO AUDIO)
TA-SVID-DIGT	DIGITAL VIDEO (NO AUDIO)
TA-SVID-EQPM	VIDEO EQUIPMENT
TA-TVVS-CABLE	COMBINED VIDEO AND AUDIO COPPER CABLES
TA-TVVS-EQPM	EQUIPMENT PROCESSESING COMBINED VIDEO AND AUDIO SIGNALS
TA-TVVS-FBMM	COMBINED VIDEO AND AUDIO MULTI MODE FIBER
TA-TVVS-FBSM	COMBINED VIDEO AND AUDIO SINGLE MODE FIBER CABLES
TA-TVVS-HDMI	DIGITAL A/V CABLES - HDMI, DVI, OR DP (VIDEO AND AUDIO COMBINED)
TN-CNXX	CENTRIXS-K (KOREA)
TN-CNXX-CABL	CENTRIXS-K COPPER CABLE
TN-CNXX-FBMM	CENTRIXS-K MULTI MODE FIBER
TN-CNXX-FBSM	CENTRIXS-K SINGLE MODE FIBER
TN-NIPR	NIPRNET (UNCLASSIFIED) NETWORK
TN-NIPR-CABL	NIPRNET COPPER CABLES
TN-NIPR-FBMM	NIPRNET MULTI MODE FIBER
TN-NIPR-FBSM	NIPRNET SINGLE MODE FIBER
TN-SIPR	SIPRNET (CLASSIFIED) NETWORK
TN-SIPR-CABL	SIPRNET COPPER CABLE
TN-SIPR-EQPM	SIPRNET EQUIPMENT (EXCLUSIVE TO SIPRNET)
TN-SIPR-FBMM	SIPRNET MULTI MODE FIBER
TN-SIPR-FBSM	SIPRNET SINGLE MODE FIBER
TN-TVVS-CTRL	A/V CONTROL NETWORK (ETHERNET)

NOTE: The A/V control layers include TN-TVVS-CTRL (or TN-CTRL) and TA-CTRL in order to differentiate between links using Ethernet and other type of digital control protocols. This could also be accomplished using layers TA-CTRL-DIGT and TA-CTRL-NTWK.

DVI=Digital Visual Interface; HDMI=High Definition Multimedia Interface

Note: A/V VIS STD drawing provided as separate AutoCAD file.



The USAISEC STD-MS-000A is shown in Figures D-1 and D-2.



D-1

FOR OFFICIAL USE ONLY



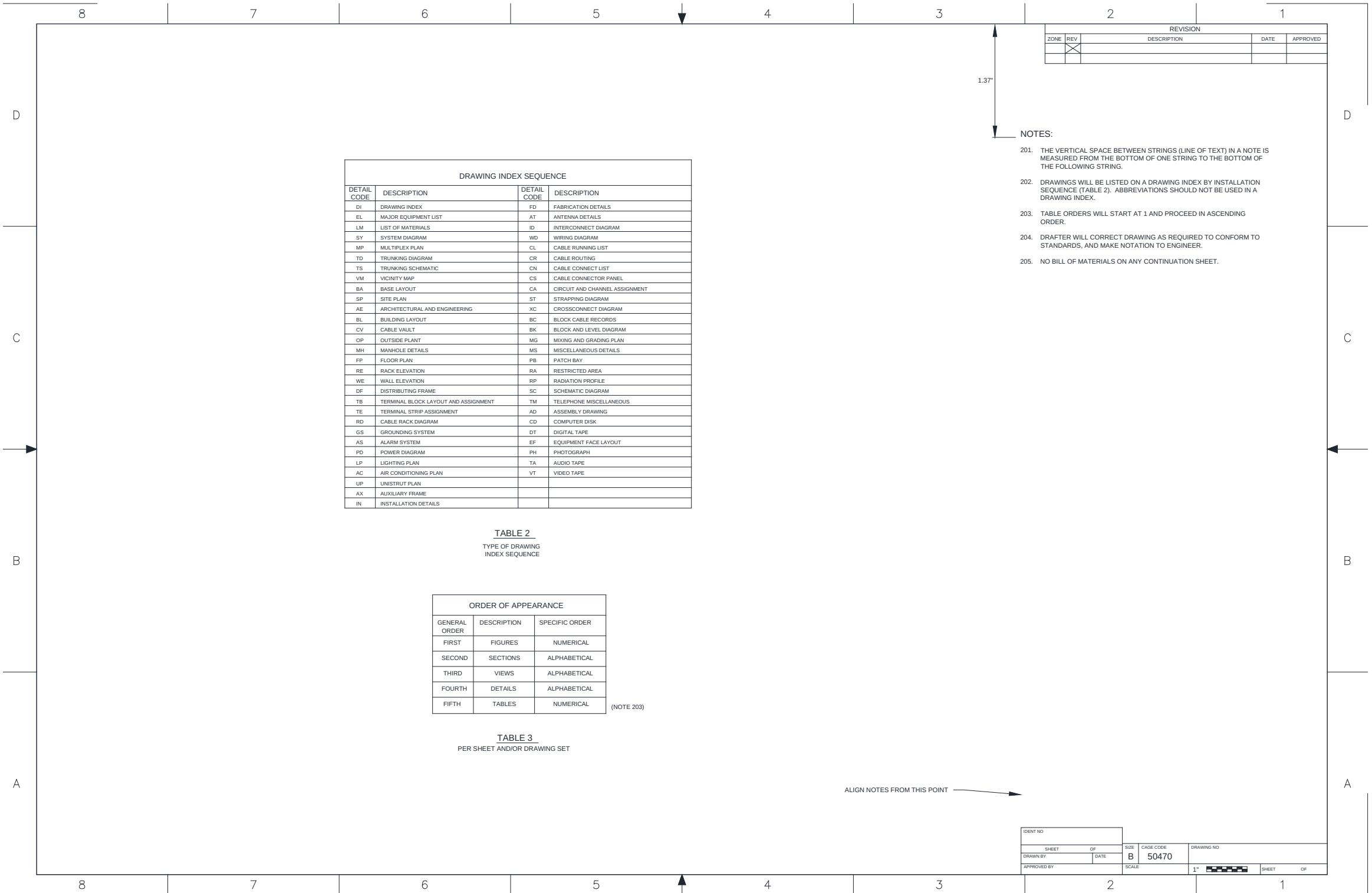


Figure D-2. STD-MS-000A Index





USAISEC Drawing Border Templates are provided in Figures E-1, E-2, and E-3.

[illegible]

Figure E-1. Drawing Border Template - Sheet 1

E-1

FOR OFFICIAL USE ONLY



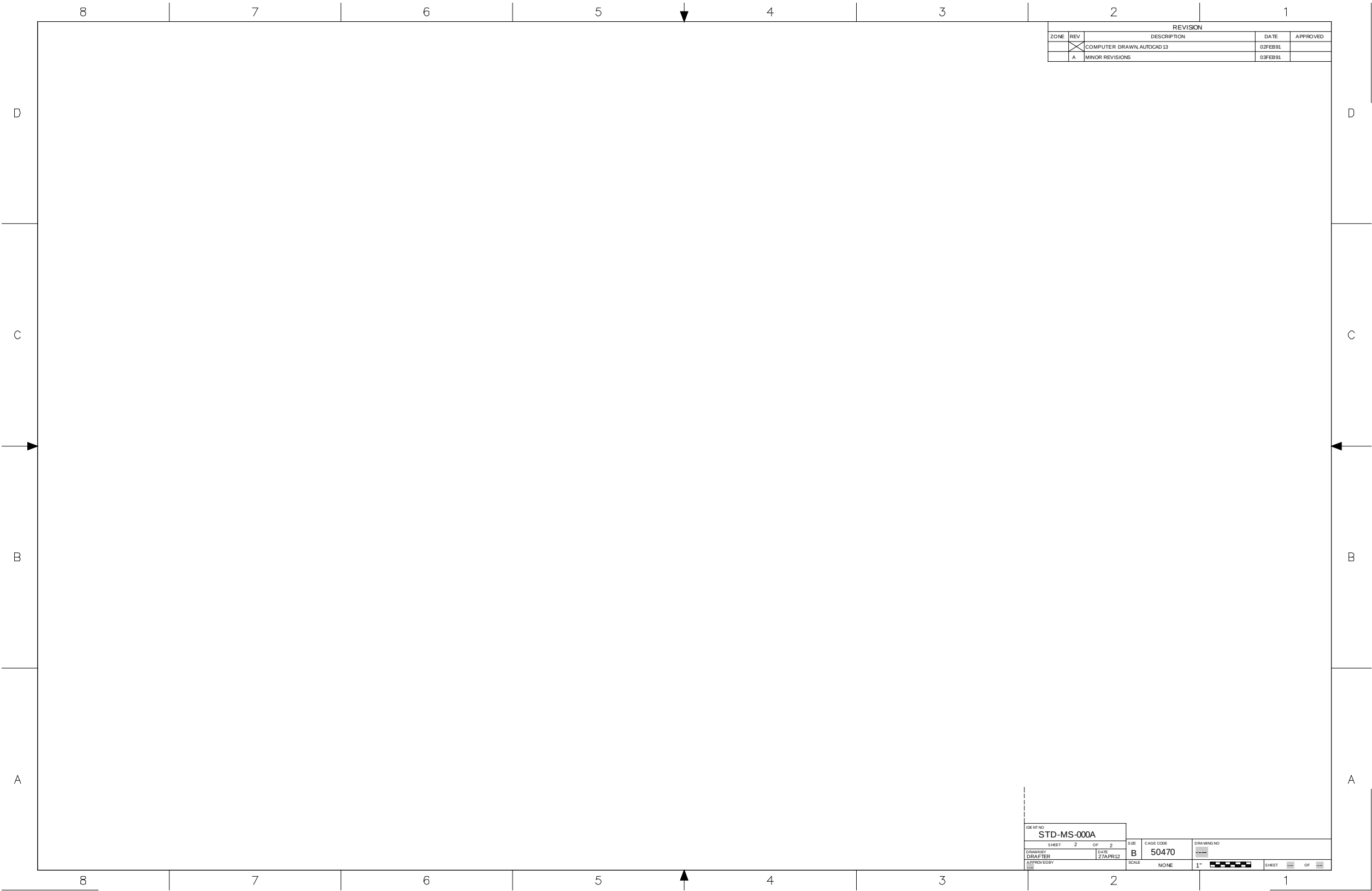


Figure E-2. Drawing Border Template - Sheet 2

E-2

FOR OFFICIAL USE ONLY





Figure E-3. Drawing Border Template - Sheet 3



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APPENDIX F. LEGACY INSTALLATION INFORMATION INFRASTRUCTURE MODERNIZATION PROGRAM (I3MP) SPECIFIC GUIDANCE

F-1.0 FILE NAMING CONVENTIONS.

The I3MP has a specific electronic file naming convention which follows the format described in the following example:

A Dense Wave-Division Multiplex (DWDM) System Diagram Drawing will use the file naming convention, `sitename_SY_O_SH_O-DWDM.dwg`. In this format, "sitename" is the three-letter I3MP site designation for the site from Table C-1, "SY" is the type of drawing (system diagram), and the first "O" is the drawing series number. The "SH" is the separator for the sheet number, the second "O" is the sheet number of the series drawing, and "DWDM" is the type of system diagram. For example, `GRN_SY_I_SH_2-DWDM.dwg` would be the file name for the System Diagram Drawing 1, Sheet 2 at Fort Gordon.

F-2.0 DRAWING NAMING CONVENTIONS

F-2.1 Title Blocks

All drawings will have title blocks. For a series of drawings, the sheet one and sheet two title blocks are different. Information describing the drawings will be entered in the drawing title blocks. Below are the types of information that will be entered in the title block/drawing formats as referenced in Appendix D, USAISEC STD-MS-000A. The numbers in the subparagraphs below correspond to the numbers on the title block examples listed in Appendix C.

F-2.2 Type of Drawing (1)

This area is used to describe the kind of drawing created. This field can have one to three lines and the first line will be the type of drawing.

The drawing types are listed as follows:

- Outside Plant
- Maintenance Hole Details
- Installation Details
- System Diagram
- Rack Elevation
- Floor Plan
- Building Layout
- Site Plan
- Block Diagram
- Wiring Diagram
- Elevation Drawing
- Drawing Notes





- Distribution Frame

Please note that the example is for sheet one drawings only. For continuation sheets, such as sheet 2 of 5, the first line of the type of drawing from sheet one will be entered above the drawing number as shown in the title block example drawings.

F-2.3 Drawing Number (2)

The USG Drafting Lead normally issues the drawing number. Drawing numbers consist of the format, 00000L-MH00001. In this format, "00000" is the installation identification number or project description number, "TL" is the type of drawing package (telephone), "MH" is the type of drawing series (Maintenance Hole), and "00001" is the sequential drawing series number.

F-3.0 MODEL LAYER NAMING CONVENTION

Table F-1 shows the I3MP Layer Naming Convention.

Table F-1. I3MP Layer Naming Convention

Task#	Layer Name	Layer Color	Color Variant	Description
1	TASK_1	1	Red	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 1
1	TASK_1_TXT	10	Red	Layer for text, descriptors, leaders, location work bubbles for Task 1
2	TASK_2	82	Green	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 2
2	TASK_2_TXT	72	Green	Layer for text, descriptors, leaders, location work bubbles for Task 2
3	TASK_3	132	Cyan	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 3
3	TASK_3_TXT	122	Cyan	Layer for text, descriptors, leaders, location work bubbles for Task 3
4	TASK_4	5	Blue	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 4
4	TASK_4_TXT	160	Blue	Layer for text, descriptors, leaders, location work bubbles for Task 4
5	TASK_5	6	Magenta	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 5
5	TASK_5_TXT	210	Magenta	Layer for text, descriptors, leaders, location work bubbles for Task 5
6	TASK_6	40	Orange	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 6
6	TASK_6_TXT	42	Orange	Layer for text, descriptors, leaders, location work bubbles for Task 6
7	TASK_7	54	Yellow	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 7
7	TASK_7_TXT	52	Yellow	Layer for text, descriptors, leaders, location work bubbles for Task 7
8	TASK_8	134	Cyan	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 8
8	TASK_8_TXT	136	Cyan	Layer for text, descriptors, leaders, location work bubbles for Task 8
9	TASK_9	140	Blue	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 9
9	TASK_9_TXT	142	Blue	Layer for text, descriptors, leaders, location work bubbles for Task 9
10	TASK_10	190	Magenta	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 10
10	TASK_10_TXT	192	Magenta	Layer for text, descriptors, leaders, location work bubbles for Task 10
11	TASK_11	244	Red	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 11
11	TASK_11_TXT	246	Red	Layer for text, descriptors, leaders, location work bubbles for Task 11

**Table F-1. I3MP Layer Naming Convention (continued)**

Task#	Layer Name	Layer Color	Color Variant	Description
12	TASK_12	44	Yellow	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 12
12	TASK_12_TXT	46	Yellow	Layer for text, descriptors, leaders, location work bubbles for Task 12
13	TASK_13	97	Green	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 13
13	TASK_13_TXT	107	Green	Layer for text, descriptors, leaders, location work bubbles for Task 13
14	TASK_14	133	Cyan	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 14
14	TASK_14_TXT	131	Cyan	Layer for text, descriptors, leaders, location work bubbles for Task 14
15	TASK_15	163	Blue	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 15
15	TASK_15_TXT	173	Blue	Layer for text, descriptors, leaders, location work bubbles for Task 15
16	TASK_16	211	Magenta	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 16
16	TASK_16_TXT	221	Magenta	Layer for text, descriptors, leaders, location work bubbles for Task 16
17	TASK_17	13	Red	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 17
17	TASK_17_TXT	11	Red	Layer for text, descriptors, leaders, location work bubbles for Task 17
18	TASK_18	43	Yellow	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 18
18	TASK_18_TXT	53	Yellow	Layer for text, descriptors, leaders, location work bubbles for Task 18
19	TASK_19	75	Green	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 19
19	TASK_19_TXT	85	Green	Layer for text, descriptors, leaders, location work bubbles for Task 19
20	TASK_20	125	Cyan	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 20
20	TASK_20_TXT	135	Cyan	Layer for text, descriptors, leaders, location work bubbles for Task 20
21	TASK_21	144	Blue	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 21
21	TASK_21_TXT	142	Blue	Layer for text, descriptors, leaders, location work bubbles for Task 21
22	TASK_22	183	Magenta	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 22
22	TASK_22_TXT	181	Magenta	Layer for text, descriptors, leaders, location work bubbles for Task 22
23	TASK_23	24	Red	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 23
23	TASK_23_TXT	14	Red	Layer for text, descriptors, leaders, location work bubbles for Task 23





Table F-1. I3MP Layer Naming Convention (continued)

Task#	Layer Name	Layer Color	Color Variant	Description
24	TASK_24	66	Green	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 24
24	TASK_24_TXT	76	Green	Layer for text, descriptors, leaders, location work bubbles for Task 24
25	TASK_23	216	Magenta	Layer for maintenance holes, hand holes, pedestals, conduits/cables for Task 25
25	TASK_23_TXT	214	Magenta	Layer for text, descriptors, leaders, location work bubbles for Task 25
	I3MP_NEW	11	Red	Layer showing lines for new I3MP engineering
	I3MP_NEW_TXT	30	Orange	Text layer for new I3MP engineering
	CAT6	201	Magenta	Layer showing lines representing Category 6 (CAT6) cable
	CAT6_TXT	200	Magenta	Text layer for CAT6 cable
	POWER	10	Red	Layer showing lines representing power paths and panels
	POWER_TXT	20	Red	Text layer for power paths and panels
	GROUND	132	Green	Layer showing lines and symbols representing grounding
	GROUND_TXT	122	Green	Text layer for grounding
	Exs_MHDuct	11	Red	Layer showing lines between maintenance holes, hand holes, pedestals and poles, depicting duct
	Exs_MHDuct_Txt	30	Orange	Text layer for duct sizes, duct callouts
	Exs_Manhole	22	Red	Layer showing maintenance hole polygons depicting maintenance hole locations
	Exs_Manhole_Txt	30	Orange	Text layers for maintenance hole numbers
	Exs_HCDS	150	Blue	Layer showing lines between hardened carrier maintenance holes, hand holes, pedestals, and poles depicting duct
	Exs_HCDS_Txt	152	Blue	Text layer for hardened carrier duct sizes, duct and cable callouts and layers depicting HCDS
	Exs_Handhole	22	Red	Layer showing hand hole polygons depicting hand hole locations
	Exs_Handhole_Txt	30	Orange	Text layer for hand hole numbers
	Exs_Copper	170	Blue	Layer showing copper terminal symbology
	Exs_Copper_Txt	180	Blue	Text layer for copper callouts and text for copper terminals
	Exs_Fiber	94	Green	Layer showing fiber terminal symbology
	Exs_Fiber_Txt	96	Green	Text layer for fiber callouts and text for fiber terminals
	Exs_Footages	31	Orange	Text layer for footages (distances between maintenance holes, hand holes, poles and pedestals)
	Exs_Pedestals	3	Green	Layer showing pedestal polygons with "x" inside depicting pedestal locations
	Exs_Pedestal_Txt	4	Cyan	Text layer for pedestal numbers
	Exs_Pole	163	Blue	Layer showing pole circles depicting pole locations
	Exs_Pole_Txt	30	Orange	Text layer for pole numbers
	Leaders	30	Orange	Layer showing arrow and hook leaders as well as any dimensions



F-4.0 DESCRIPTOR BLOCKS

In most cases, descriptor blocks are inserted within the OSP and Installation Detail Drawings of an I3MP engineering package. Drawing descriptor blocks will be inserted into the drawings on their appropriate layer. For example, all work for Task 1 of a project will be drawn on the Task I Layer. All text associated with Task 1 of a project, to include the standard drawing descriptor blocks, will be drawn or inserted on the Task_1_Txt Layer with the appropriate color (see Appendix B, Transmission System Engineering Drawings). Table F-2 identifies the standard three-letter site designations for I3MP. Table F-3 identifies the Legacy I3MP Standard Drawing Symbols.

a. Work Location. This is a sequential numeric value identifying the location of the work with the associated descriptors to be performed on an engineering drawing.

b. Quantity. This is a numeric value identifying how many of each descriptor item is needed at a specific work location.

c. Remarks. Any text value required to add further information to the work location actions and/or descriptor blocks.

d. Funded. This is a text value defining who will be funding the work at a specific work location.

Table F-2 identifies the three-letter site description for I3MP.

Table F-2. Standard Three-Letter Site Designations for I3MP

Location Code	Location		Location Code	Location
ABB	ABBEVILLE, AL		DNX	FT KNOX, KY
ABQ	ALBUQUERQUE, NM		DPG	DUGWAY PROVING GROUNDS, UT
ABR	AUBROOK, CZ		DRM	FT DRUM, NY
ADE	ADELPHI, MD		DVN	DAVISON AAF, VA
AGS	AUGUSTA, ME		EDG	EDGEWOOD ARSENAL, MD
AHS	ARLINGTON HALL STA, VA		EUS	FT EUSTIS, VA
AIA	ABROOK INDUSTRIAL AREA		FGI	FT GREELY, AK
AJC	SITE R FT RITCHIE, MD		FMC	FT McCLELLAN, AL
AKX	ALASKA		FMN	FT MONROE, VA
ALB	ALBANY, NY		FMT	FT MONMOUTH, NJ
ALE	ALEXANDRIA, VA		FRN	FT RICHARDSON, AK
ALL	FT ALLEN, PR		FSH	FT SHAFTER, HI
ALX	ALABAMA		FTD	FT DETRICK, MD
AMC	AMC HQ		FTH	FT HUACHUCA, AZ
AND	ANDREWS AFB		FTL	FT LEAVENWORTH, KS
ANN	ANNISTON AD, AL		FTW	FT WAINWRIGHT, AK
APG	ABERDEEN PROVING GROUND, MD		FWG	FT WINGATE, NM
APH	FT AP HILL, VA		GEO	CAMP GEORGE, KOREA



**Table F-2. Standard Three-Letter Site Designations for I3MP (continued)**

Location Code	Location		Location Code	Location
ARL	ARLINGTON, VA		GET	GRAF EAST TRAINING CAMP, FRG
ARP	ARLINGTON POINT, AL		GIE	GIEBELSTADT, FRG
ARX	ARKANSAS		GIL	FT GILLEM, GA
ATA	ATLANTA AD, GA		GRA	GRAFENWOEHR, FRG
ATH	ATHENS, GA		GRN	FT GORDON, GA
ATL	ATLANTA, GA		HAF	HUNTER AAF, GA
ATT	CAMP ATTERBURY, IN		HAN	HANAU, FRG
AUG	ST AUGUSTINE, FL		HAP	HOLSTON AAP, IN
AUS	AUSTIN, TX		HAT	HAWTHORNE NAVAL AMMUNITION DEPOT
AZX	ARIZONA		HER	HERNDON, VA
BAG	BADGERAAP, WI		HEN	CAMP HENRY, KOREA
BAL	BALTIMORE, MD		HMN	FT HAMILTON, KY
BAT	BATTLE CREEK, MI		HOG	HOHENFELS, FRG
BAY	BAYONNE MIL OCEAN TML, NJ		HOO	FT HOOD, TX
BEL	FT BELVOIR, VA		HUM	CAMP HUMPHREYS, KOREA
BEN	FT BENNING, GA		IRW	FT IRWIN, CA
BHA	FT BEN HARRISON, IN		JAG	FT JACKSON, SC
BIS	BISMARCK, NO		KAN	KANSAS CITY, MO
BKE	BROOKE AMC, TX		KIT	KITZINGEN, FRG
BKH	BUCKHANNON, WV		LBG	LEIGHTON BARRACKS, FRG
BKR	FT BAKER, CA		LEE	FT LEE, VA
BKY	BROOKLYN, NY		LET	LETTERKENNY AD, PA
BLI	FT BLISS, TX		LEX	LEXINGTON BLUE GRASS AD, KY
BOI	BOISE 10		LWD	FT LEONARD WOOD, MO
BOO	BROOKLYN AR TML, NY		LWS	FT LEWIS, WA
BOS	BOSTON, MA		MCN	FT McNAIR, DC
BOT	BOTHELL, WA		MOE	FT GEORGE G. MEADE, MD
BPR	BATTERY PRATT, CZ		MPH	FT McPHERSON, GA
BRA	FT BRAGG, NC		MYR	FT MYER, VA
BRH	BROOKHAVEN, MS		ORO	FT ORD, CA
BRY	FT BARRY, CA		PAD	PUEBLO AD, CO
BUG	FT BUCHANAN, PR		PBA	PINE BLUFF ARSENAL, AR
BXR	BAILEYS CROSSROADS, VA		PBG	PATCH BARRACKS, FRG
CAR	CAMP CARROLL, KOREA		PIC	PICATINNY ARSENAL, NJ
CAS	CAMP CASTLE, KOREA		PKT	FT PICKETT, VA
CAX	CALIFORNIA		PLK	FT POLK, LA
CBJ	CAMP BUCKNER, JAPAN		PNT	PENTAGON, VA
CBK	CARLISLE BARRACKS, PA		RED	REDSTONE ARSENAL, AL
CBL	FT CAMPBELL, KY		RIA	ROCK ISLAND ARSENAL, IL



**Table F-2. Standard Three-Letter Site Designations for I3MP (continued)**

Location Code	Location		Location Code	Location
CCI	CORPUS CHRISTI, TX		RKR	FT RUCKER, AL
CCK	GARNO CASEV, KOREA		RLY	FT RILEY, KS
COL	CAMP DOUGLAS, WI		ROB	RHINE ORDNANCE BARRACKS, FRG
CEI	CAMP EDERLE, ITALY		RRA	RED RIVER AD, TX
CHA	CHARLESTON, WV		SCH	SCHWEINFURT, FRG
CHF	FT CHAFFEE, AK		SLL	FT SILL, OK
CHI	CHIVA CHIVA, CZ		SMH	FT SAM HOUSTON, TX
CHK	CAMP HOVEY, KOREA		STE	FT STEWARD, GA
GHS	CHARLESTON, SC		STY	FT STORY, VA
CID	COLUMBIA PIKE ARLINGTON, VA		SUN	SUNNYPPOINT MIL OC TML, NC
CMK	CAMP MOBILE, KOREA		TBY	TOBYHANNA AD, PA





Table F-3. Legacy I3MP Standard Drawing Symbols

BLOCK	BLOCK NAME	DESCRIPTION
	I3MP_CO_TERM_L.dwg	PROPOSED COPPER BUILDING TERMINAL FOR OSPDRAWINGS(LEFT)
	I3MP_CO_TERM_R.dwg	PROPOSED COPPER BUILDING TERMINAL FOR OSPDRAWINGS(RIGHT)
	I3MP_FO_TERM_L.dwg	PROPOSED FIBER BUILDING TERMINAL FOR OSPDRAWINGS(LEFT)
	I3MP_FO_TERM_R.dwg	PROPOSED FIBER BUILDING TERMINAL FOR OSPDRAWINGS(RIGHT)
	I3MP_MH.dwg	PROPOSED MANHOLE FOR OSPDRAWINGS
	I3MP_HH.dwg	FOR OSPDRAWINGS PROPOSED HANDHOLE
	I3MP_PED.dwg	PROPOSED PEDESTAL FOR OSPDRAWINGS
	I3MP_PIPE.dwg	PROPOSED GIP OR GSP FOR OSPDRAWINGS
	I3MP_GUY.dwg	PROPOSED GUY WIRE AND ANCHOR FOR POLES ON OSP DRAWINGS
	I3MP_WL.dwg	WORK LOCATION BUBBLE AND NUMBER FOR OSPDRAWINGS
	I3MP_ADN.dwg	SYMBOL DENOTING THE LOCATION OF AN ADN OR MCN ON OSP DRAWINGS (THIS WILL BE PLACED BESIDE OF THE BLDG THAT ACTS AS AN ADN OR MCN AND WILL BE ON THE APPROPRIATE LAYER PER LAYER STDS)

**Table F-3. Legacy I3MP Standard Drawing Symbols (continued)**

BLOCK	BLOCK NAME	DESCRIPTION
	I3MP_6DUCT.dwg	PROPOSED 6 WAY DUCT BANK (I3MP_2DUCT.DWG WOULD BE A 2 WAY DUCT) (I3MP_4DUCT.DWG WOULD BE A 4 WAY DUCT) (I3MP_9DUCT.DWG WOULD BE A 9 WAY DUCT) (I3MP_12DUCT.DWG WOULD BE A 12 WAY DUCT)
	I3MP_4ID.dwg	INNERDUCT FOR DUCT BANKS LISTED ABOVE (INSERT POINT IS CENTER OF THE DUCT)
	I3MP_SUB.dwg	SUB-DUCT FOR DUCT BANKS LISTED ABOVE (INSERT POINT IS CENTER OF THE DUCT)
	I3MP_MH_RACK.dwg	PROPOSED RACKING FOR MANHOLES
	I3MP_CO_SPL.dwg	PROPOSED COPPER SPLICE CASE FOR MANHOLE DETAILS
	I3MP_FO_SPL.dwg	PROPOSED FIBER SPLICE CASE FOR MANHOLE DETAILS
	I3MP_FO_LOOP.dwg	PROPOSED FIBER OPTIC SERVICE LOOP FOR MANHOLE DETAILS
	I3MP_GND.dwg	PROPOSED GROUND ROD FOR OSP MANHOLE DETAILS





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GLOSSARY. ACRONYMS AND ABBREVIATIONS

+	Plus
A/E/C	Architectural, Engineering, and Construction
A/V	Audiovisual
AC	Alternating Current
AE	Architectural and Engineering Drawing
ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
AZ	Arizona
BA	Base Layout Drawing
BE	Belgium
BL	Building Layout Drawing
C	CONUS
C-E	Communications-Electronics
CA	Circuit and Channel Assignment Drawing
CAD	Computer Aided Design
CAT6	Category 6
CATV	Cable Television
CDF	Combined Distribution Frame
CENTRIXS	Combined Enterprise Regional Information Exchange System
CENTRIXS-K	CENTRIXS-Korea
CL	Cable Running List
CONUS	Continental United States
CRL	Cable Running List
CV	Cable Vault Drawing
DC	Direct Current
DF	Distribution Frame/Distribution Frame Drawing
DI	Drawing Index
DoD	Department of Defense
DVI	Digital Visual Interface
DWDM	Dense Wave-Division Multiplex
E	Europe
EES	Earth Electrode Subsystem
EIP	Engineering Installation Plan
EMP	Electromagnetic Pulse
FOC	Fiber Optic Cable
FP	Floor Plan
FT	Feet
FTH	Fort Huachuca

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GE	Germany
GS	Grounding System Drawing
HDMI	High Definition Multimedia Interface
HVAC	Heating, Ventilation, and Air Conditioning
I3MP	Installation Information Infrastructure Modernization Program
ID	Identification/Interconnect Diagram
IDF	Intermediate Distribution Frame
IN	Installation Drawing
ISO	International Organization for Standardization
KO	Korea
LOM	List of Materials
MDF	Main Distribution Frame
MET	Modernization of Enterprise Terminals
MH	Maintenance Hole Detail Drawing
MH	Maintenance Hole
mm	millimeter
MM	Multimode
MS	Miscellaneous Detail Drawing
MW	Microwave
N/A	Not Applicable
NCS	National CAD Standard
NIPRNet	Non-classified Internet Protocol Router Network
NSN	National Stock Number
OP	Outside Plant Drawing
OSP	Outside Plant
P	Pacific
PD	Power Diagram
PDC	Primary Domain Controller
PL	Project Lead
PM	Program Manager
QTY	Quantity
RD	Cable Rack Diagram
RDECOM	Research, Development, and Engineering Communications
RE	Rack Elevation Drawing
REV	Revision





SCADA	Supervisory Control and Data Acquisition
SD	Standard Drawing
SIPRNet	Secret Internet Protocol Router
SM	Single Mode
SP	Site Plan
STD	Standard/Standard Drawing
SWA	Southwest Asia
SY	System Diagram
TA	A/V Drawing
TB	Terminal Block Layout and Assignment Drawing
TC	Technical Control
TD	Technical Director
TL	Telephone Switch
TTC	Terrestrial Technical Control
TYAD	Tobyhanna Army Depot
UI	Unit of Issue
UP	Unistrut Plan
UPS	Uninterruptible Power Supply
USACE	U.S. Army Corps of Engineers
USAISEC	U.S. Army Information Systems Engineering Command
USG	U.S. Government
VIA	Patch Barracks
VIS	Visual Information System
WE	Wall Elevation Drawing
XC	Cross-connect Diagram





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